# T3 UV one-loop RGE term comparison

## Notation key

| symbol | interaction / definition |
|---|---|
| $g_1$ | $D_\mu \supset i g_1 Y B_\mu$ |
| $g_2$ | $D_\mu \supset i g_2 T^A W_\mu^A$ |
| $g_3$ | $D_\mu \supset i g_3 t^A G_\mu^A$ |
| $Y_u$ | $-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$ |
| $Y_d$ | $-\bar{Q} Y_d H d + \text{h.c.}$ |
| $Y_e$ | $-\bar{L} Y_e H e + \text{h.c.}$ |
| $y_1$ | $L F S_1 + \text{h.c.}$ |
| $y_2$ | $L F S_2 + \text{h.c.}$ |
| $M_F$ | $-\bar{F} M_F F$ |
| $T$ | $\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$ | $\text{Tr}(y_1^\dagger y_1)$ |
| $T_\nu^{(2)}$ | $\text{Tr}(y_2^\dagger y_2)$ |
| $T_\nu$ | $T_\nu^{(1)} + T_\nu^{(2)}$ |
| $m_1^2$ | $m_1^2 S_1^\dagger S_1$ |
| $m_2^2$ | $m_2^2 S_2^\dagger S_2$ |
| $\lambda_1$ | $\frac{1}{2}\lambda_1 (H^\dagger H)^2$ |
| $\lambda_{S_1}^{(1)}$ | $\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$ |
| $\lambda_{S_2}^{(1)}$ | $\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$ |
| $\lambda_{HS_1}^{(1)}$ | $\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$ |
| $\lambda_{HS_2}^{(1)}$ | $\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$ |
| $\lambda_{HS_1}^{(A)}$ | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$ |
| $\lambda_{HS_2}^{(A)}$ | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$ |
| $\lambda_5$ | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$ |
| $\lambda_{12}^{(1)}$ | $\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$ |
| $\lambda_{12}^{(A)}$ | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$ |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$ |
| $\lambda_{S_1}^{(A)}$ | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$ |
| $\lambda_{S_2}^{(A)}$ | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$ |
| $\alpha$ | $Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$ |
| $C_2(d)$ | $(d^2-1)/4$ |
| $T(d)$ | $d(d^2-1)/12$ |
| $C_{LLS_1S_2}^{ij;AB}$ | $L_i L_j S_1^A S_2^B$ |
| $C_{ijab}$ | $L_i L_j \phi_a \phi_b$ |
| $C_5^{ij}$ | $(L_i^T C L_j) H H$ |

| | |
|---|---|
| $R_W$ | $16\pi^2\beta_{C_5} \supset R_W\lambda_5 C_{LLS_1S_2}$ |

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2\,\mu\,dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \to g$ conversion as in the previous report.

$\beta_{g_1}^{(1)}$

**Quantity:** $g_1$

$$16\pi^2\,\mu\frac{dg_1}{d\mu} = \beta_{g_1}^{(1)}$$

| Column | Representative term structure |
|---|---|
| T1 | $\frac{163{g_1}^3}{6}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\frac{163{g_1}^3}{6}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\frac{55{g_1}^3}{6}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\frac{59{g_1}^3}{6}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $\frac{25{g_1}^3}{2}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\frac{43{g_1}^3}{6}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $\frac{25{g_1}^3}{2}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\frac{41{g_1}^3}{2}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $\frac{43{g_1}^3}{6}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{41{g_1}^3}{2}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $\frac{59{g_1}^3}{6}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $\frac{55{g_1}^3}{6}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $\frac{163{g_1}^3}{6}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\frac{179{g_1}^3}{6}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\frac{59{g_1}^3}{6}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\frac{59{g_1}^3}{6}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $\frac{179{g_1}^3}{6}$ |

$\beta_{g_2}^{(1)}$

**Quantity:** $g_2$

$$16\pi^2\,\mu\frac{dg_2}{d\mu} = \beta_{g_2}^{(1)}$$

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{{g_2}^3}{2}$ |
| T2 | $-\frac{17{g_2}^3}{6}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{{g_2}^3}{2}$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{{g_2}^3}{2}$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{{g_2}^3}{2}$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $-\frac{17{g_2}^3}{6}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $-\frac{17{g_2}^3}{6}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $-\frac{17{g_2}^3}{6}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $\frac{31{g_2}^3}{6}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $\frac{7{g_2}^3}{6}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $\frac{31{g_2}^3}{6}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{{g_2}^3}{2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $-\frac{{g_2}^3}{2}$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $-\frac{{g_2}^3}{2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\frac{{g_2}^3}{6}$ | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\frac{{g_2}^3}{6}$ | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\frac{{g_2}^3}{6}$ | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $\frac{{g_2}^3}{6}$ | — |

$\beta_{g_3}^{(1)}$

**Quantity:** $g_3$

$$16\pi^2\,\mu\frac{dg_3}{d\mu} = \beta_{g_3}^{(1)}$$

| Column | Representative term structure |
|---|---|
| T1 | $-7{g_3}^3$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-7{g_3}^3$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-7{g_3}^3$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-7{g_3}^3$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-7{g_3}^3$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-7{g_3}^3$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-7{g_3}^3$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-7{g_3}^3$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-7{g_3}^3$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-7{g_3}^3$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $-7{g_3}^3$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $-7{g_3}^3$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $-7{g_3}^3$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-7{g_3}^3$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-7{g_3}^3$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-7{g_3}^3$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $-7{g_3}^3$ |

$\beta^{(1)}_{Y_u}$

**Quantity:** $Y_u$

$$16\pi^2\,\mu\frac{dY_u}{d\mu} = \beta^{(1)}_{Y_u}$$

## Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ |
| T2 | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ |
| T3 | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_u)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_u)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_u)_{ij}$ |

**Quantity:** $Y_u$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ |
| T2 | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ |
| T3 | $-8{g_3}^2\ (Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_u)_{ij}$ | $-8{g_3}^2\ (Y_u)_{ij}$ |

**Quantity:** $Y_u$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $-\frac{17}{12} {g_1}^2 (Y_u)_{ij}$ |

$\beta^{(1)}_{Y_d}$

**Quantity:** $Y_d$

$$16\pi^2\,\mu\frac{dY_d}{d\mu} = \beta^{(1)}_{Y_d}$$

## Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ |
| T2 | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ |
| T3 | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_d)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_d)_{ij}$ |

**Quantity:** $Y_d$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ |
| T2 | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ |
| T3 | $-8{g_3}^2\ (Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\ (Y_d)_{ij}$ | $-8{g_3}^2\ (Y_d)_{ij}$ |

**Quantity:** $Y_d$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $-\frac{5}{12} {g_1}^2 (Y_d)_{ij}$ |

$\beta_{Y_e}^{(1)}$

**Quantity:** $Y_e$

$$16\pi^2\,\mu\frac{dY_e}{d\mu} = \beta_{Y_e}^{(1)}$$

## Terms 1 of 3

| Column | Representative term structure |
|---|---|
| T1 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ |
| T2 | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ |
| T3 | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $3\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)(Y_e)_{ij}$ | $+3\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)(Y_e)_{ij}$ | $+\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)(Y_e)_{ij}$ |

**Quantity:** $Y_e$

# Terms 2 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij}+3\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij}+6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |
| T2 | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ |
| T3 | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij}+3\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij}+6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij}+3\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij}+6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{1}{4}(2\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij}+3\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij}+6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij}+2\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij}+6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij}+2\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij}+6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij}+2\left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij}+6\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$ |

**Quantity:** $Y_e$

# Terms 3 of 3

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 3\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\frac{3}{4}(\left(y_1\ (y_1)^\dagger\ Y_e\right)_{ij} + \left(y_2\ (y_2)^\dagger\ Y_e\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij})$ |

$\beta^{(1)}_{y_1}$

**Quantity:** $y_1$

$$16\pi^2\,\mu\frac{dy_1}{d\mu} = \beta^{(1)}_{y_1}$$

## Terms 1 of 5

| Column | Representative term structure |
|---|---|
| T1 | $2T^{(1)}_{\nu}\,(y_1)_{ij}$ |
| T2 | $+\frac{3}{4}\left(y_2\;(y_2)^\dagger\;y_1\right)_{ij}$ |
| T3 | $+\frac{1}{2}\left(Y_e\;(Y_e)^\dagger\;y_1\right)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $2T^{(1)}_{\nu}\,(y_1)_{ij}$ | $+\frac{3}{4}\left(y_2\;(y_2)^\dagger\;y_1\right)_{ij}$ | $+\frac{1}{2}\left(Y_e\;(Y_e)^\dagger\;y_1\right)_{ij}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $2T^{(1)}_{\nu}\,(y_1)_{ij}$ | $+\frac{3}{4}\left(y_2\;(y_2)^\dagger\;y_1\right)_{ij}$ | $+\frac{1}{2}\left(Y_e\;(Y_e)^\dagger\;y_1\right)_{ij}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $2T^{(1)}_{\nu}\,(y_1)_{ij}$ | $+\frac{3}{4}\left(y_2\;(y_2)^\dagger\;y_1\right)_{ij}$ | $+\frac{1}{2}\left(Y_e\;(Y_e)^\dagger\;y_1\right)_{ij}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $y_1$

# Terms 2 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij}$ |
| T2 | $-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$ |
| T3 | $-\frac{15}{2}{g_1}^2\,(y_1)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij}$ | $-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$ | $-\frac{15}{2}{g_1}^2\,(y_1)_{ij}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij}$ | $-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$ | $-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij}$ | $-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$ | $-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $-\frac{9}{4}{g_2}^2\,(y_1)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(y_1)_{ij}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $-\frac{9}{4}{g_2}^2\,(y_1)_{ij}$ | $-\frac{3}{4}{g_1}^2\,(y_1)_{ij}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $-\frac{9}{4}{g_2}^2\,(y_1)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(y_1)_{ij}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $-\frac{33}{4}{g_2}^2\,(y_1)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(y_1)_{ij}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $-\frac{33}{4}{g_2}^2\,(y_1)_{ij}$ | $-\frac{3}{4}{g_1}^2\,(y_1)_{ij}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $-\frac{33}{4}{g_2}^2\,(y_1)_{ij}$ | $-\frac{15}{4}{g_1}^2\,(y_1)_{ij}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$ | $-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$ | $-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$ | $-\frac{15}{2}{g_1}^2\,(y_1)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$ | $-\frac{15}{2}{g_1}^2\,(y_1)_{ij}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$ | $-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$ | $-\frac{3}{2}{g_1}^2\,(y_1)_{ij}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $-\frac{9}{2}{g_2}^2\,(y_1)_{ij}$ | $-\frac{15}{2}{g_1}^2\,(y_1)_{ij}$ |

**Quantity:** $y_1$

# Terms 3 of 5

| Column | Representative term structure |
|---|---|
| T1 | $T_\nu^{(1)}\,(y_1)_{ij}$ |
| T2 | $+\frac{1}{2}(\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ |
| T3 | $\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)(y_2)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $T_\nu^{(1)}\,(y_1)_{ij}$ | $+\frac{1}{2}(\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+T_\nu^{(1)}\,(y_1)_{ij}$ | — | $\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)(y_2)_{ij}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $T_\nu^{(1)}\,(y_1)_{ij}$ | $+\frac{1}{2}(\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $T_\nu^{(1)}\,(y_1)_{ij}$ | $+\frac{1}{2}(\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $T_\nu^{(1)}\,(y_1)_{ij}$ | $+\frac{1}{2}(\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $T_\nu^{(1)}\,(y_1)_{ij}$ | $+\frac{1}{2}(\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij} + 3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $T_\nu^{(1)}\,(y_1)_{ij}$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $T_\nu^{(1)}\,(y_1)_{ij}$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $T_\nu^{(1)}\,(y_1)_{ij}$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $T_\nu^{(1)}\,(y_1)_{ij}$ | — | — |

**Quantity:** $y_1$

# Terms 4 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{1}{2}(2\left(y_1\ (y_2)^\dagger\ y_2\right)_{ij}+\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ |
| T2 | $\frac{3}{2}T_\nu^{(1)}\ (y_1)_{ij}$ |
| T3 | $+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+5\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\frac{1}{2}(2\left(y_1\ (y_2)^\dagger\ y_2\right)_{ij}+\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+3\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $\frac{3}{2}T_\nu^{(1)}\ (y_1)_{ij}$ | $+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+5\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $+\frac{3}{2}T_\nu^{(1)}\ (y_1)_{ij}$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $\frac{3}{2}T_\nu^{(1)}\ (y_1)_{ij}$ | $+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+5\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | $+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+6\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | $+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+6\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | $+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+6\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | $+\frac{1}{4}(3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+6\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ |

**Quantity:** $y_1$

# Terms 5 of 5

| Column | Representative term structure |
|---|---|
| T1 | $\frac{3}{2}\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)(y_2)_{ij}$ |
| T2 | $+\frac{1}{4}(2\left(y_1\ (y_2)^\dagger\ y_2\right)_{ij}+3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+5\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $\frac{3}{2}\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)(y_2)_{ij}$ | $+\frac{1}{4}(2\left(y_1\ (y_2)^\dagger\ y_2\right)_{ij}+3\left(y_2\ (y_2)^\dagger\ y_1\right)_{ij}+2\left(Y_e\ (Y_e)^\dagger\ y_1\right)_{ij}+5\left(y_1\ (y_1)^\dagger\ y_1\right)_{ij})$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — |

$\beta_{y_2}^{(1)}$

**Quantity:** $y_2$

$$16\pi^2\,\mu\frac{dy_2}{d\mu} = \beta_{y_2}^{(1)}$$

## Terms 1 of 5

| Column | Representative term structure |
|---|---|
| T1 | $T_\nu^{(2)}\,(y_2)_{ij}$ |
| T2 | $+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$ |
| T3 | $-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $T_\nu^{(2)}\,(y_2)_{ij}$ | $+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $T_\nu^{(2)}\,(y_2)_{ij}$ | $+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $T_\nu^{(2)}\,(y_2)_{ij}$ | $+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $T_\nu^{(2)}\,(y_2)_{ij}$ | $+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\,(y_2)_{ij}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+T_\nu^{(2)}\,(y_2)_{ij}$ | — | $-\frac{9}{4}{g_2}^2\,(y_2)_{ij}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $T_\nu^{(2)}\,(y_2)_{ij}$ | $+\frac{1}{2}(\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + \left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 3\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-\frac{9}{4}{g_2}^2\,(y_2)_{ij}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | $-\frac{33}{4}{g_2}^2\,(y_2)_{ij}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | $-\frac{33}{4}{g_2}^2\,(y_2)_{ij}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | $-\frac{33}{4}{g_2}^2\,(y_2)_{ij}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | $-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | $-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | $-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $T_\nu^{(2)}\,(y_2)_{ij}$ | — | $-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $T_\nu^{(2)}\,(y_2)_{ij}$ | — | $-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $T_\nu^{(2)}\,(y_2)_{ij}$ | — | $-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $T_\nu^{(2)}\,(y_2)_{ij}$ | — | $-\frac{9}{2}{g_2}^2\,(y_2)_{ij}$ |

**Quantity:** $y_2$

# Terms 2 of 5

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{15}{2} {g_1}^2 \left(y_2\right)_{ij}$ |
| T2 | $\mathrm{Tr}\left(y_2 \ \left(y_1\right)^\dagger\right) \left(y_1\right)_{ij}$ |
| T3 | $+\frac{1}{2}(\left(y_1 \ \left(y_1\right)^\dagger \ y_2\right)_{ij} + 2 \left(y_2 \ \left(y_1\right)^\dagger \ y_1\right)_{ij} + \left(Y_e \ \left(Y_e\right)^\dagger \ y_2\right)_{ij} + 3 \left(y_2 \ \left(y_2\right)^\dagger \ y_2\right)_{ij})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{15}{2} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{3}{2} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{3}{2} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-\frac{15}{4} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-\frac{3}{4} {g_1}^2 \left(y_2\right)_{ij}$ | $\mathrm{Tr}\left(y_2 \ \left(y_1\right)^\dagger\right) \left(y_1\right)_{ij}$ | $+\frac{1}{2}(\left(y_1 \ \left(y_1\right)^\dagger \ y_2\right)_{ij} + 2 \left(y_2 \ \left(y_1\right)^\dagger \ y_1\right)_{ij} + \left(Y_e \ \left(Y_e\right)^\dagger \ y_2\right)_{ij} + 3 \left(y_2 \ \left(y_2\right)^\dagger \ y_2\right)_{ij})$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-\frac{15}{4} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{15}{4} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-\frac{3}{4} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{15}{4} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{3}{2} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $-\frac{3}{2} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $-\frac{15}{2} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{15}{2} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{3}{2} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{3}{2} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $-\frac{15}{2} {g_1}^2 \left(y_2\right)_{ij}$ | — | — |

**Quantity:** $y_2$

# Terms 3 of 5

| Column | Representative term structure |
|---|---|
| T1 | $\frac{3}{2} T_\nu^{(2)} \, (y_2)_{ij}$ |
| T2 | $+\frac{1}{4}(3 \left(y_1 \, (y_1)^\dagger \, y_2\right)_{ij} + 2 \left(Y_e \, (Y_e)^\dagger \, y_2\right)_{ij} + 5 \left(y_2 \, (y_2)^\dagger \, y_2\right)_{ij})$ |
| T3 | $\frac{3}{2} \, \mathrm{Tr}\left(y_2 \, (y_1)^\dagger\right) (y_1)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\frac{3}{2} T_\nu^{(2)} \, (y_2)_{ij}$ | $+\frac{1}{4}(3 \left(y_1 \, (y_1)^\dagger \, y_2\right)_{ij} + 2 \left(Y_e \, (Y_e)^\dagger \, y_2\right)_{ij} + 5 \left(y_2 \, (y_2)^\dagger \, y_2\right)_{ij})$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{3}{2} T_\nu^{(2)} \, (y_2)_{ij}$ | — | $\frac{3}{2} \, \mathrm{Tr}\left(y_2 \, (y_1)^\dagger\right) (y_1)_{ij}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{3}{2} T_\nu^{(2)} \, (y_2)_{ij}$ | $+\frac{1}{4}(3 \left(y_1 \, (y_1)^\dagger \, y_2\right)_{ij} + 2 \left(Y_e \, (Y_e)^\dagger \, y_2\right)_{ij} + 5 \left(y_2 \, (y_2)^\dagger \, y_2\right)_{ij})$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $+\frac{1}{4}(3 \left(y_1 \, (y_1)^\dagger \, y_2\right)_{ij} + 2 \left(Y_e \, (Y_e)^\dagger \, y_2\right)_{ij} + 6 \left(y_2 \, (y_2)^\dagger \, y_2\right)_{ij})$ | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $+\frac{1}{4}(3 \left(y_1 \, (y_1)^\dagger \, y_2\right)_{ij} + 2 \left(Y_e \, (Y_e)^\dagger \, y_2\right)_{ij} + 6 \left(y_2 \, (y_2)^\dagger \, y_2\right)_{ij})$ | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $+\frac{1}{4}(3 \left(y_1 \, (y_1)^\dagger \, y_2\right)_{ij} + 2 \left(Y_e \, (Y_e)^\dagger \, y_2\right)_{ij} + 6 \left(y_2 \, (y_2)^\dagger \, y_2\right)_{ij})$ | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $+\frac{1}{4}(3 \left(y_1 \, (y_1)^\dagger \, y_2\right)_{ij} + 2 \left(Y_e \, (Y_e)^\dagger \, y_2\right)_{ij} + 6 \left(y_2 \, (y_2)^\dagger \, y_2\right)_{ij})$ | — |

**Quantity:** $y_2$

# Terms 4 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(y_2\ (y_1)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 5\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$ |
| T2 | $2T_\nu^{(2)}\ (y_2)_{ij}$ |
| T3 | $+\frac{3}{4}\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{1}{4}(3\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij} + 2\left(y_2\ (y_1)^\dagger\ y_1\right)_{ij} + 2\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij} + 5\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij})$ | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $2T_\nu^{(2)}\ (y_2)_{ij}$ | $+\frac{3}{4}\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $2T_\nu^{(2)}\ (y_2)_{ij}$ | $+\frac{3}{4}\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $2T_\nu^{(2)}\ (y_2)_{ij}$ | $+\frac{3}{4}\left(y_1\ (y_1)^\dagger\ y_2\right)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $y_2$

# Terms 5 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{1}{2}\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij}$ |
| T2 | $+\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{1}{2}\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij}$ | $+\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+\frac{1}{2}\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij}$ | $+\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+\frac{1}{2}\left(Y_e\ (Y_e)^\dagger\ y_2\right)_{ij}$ | $+\left(y_2\ (y_2)^\dagger\ y_2\right)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — |

$\beta^{(1)}_{M_F}$

**Quantity:** $M_F$

$$16\pi^2\,\mu\,\frac{dM_F}{d\mu} = \beta^{(1)}_{M_F}$$

## Terms 1 of 5

| Column | Representative term structure |
|---|---|
| T1 | $\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ |
| T2 | $-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$ |
| T3 | $-\frac{27}{2}{g_1}^2\,(M_F)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$ | $-\frac{27}{2}{g_1}^2\,(M_F)_{ij}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$ | $-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 3\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$ | $-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 2\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$ | $-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 2\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$ | $-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + 2\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$ | $-\frac{27}{2}{g_1}^2\,(M_F)_{ij}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$ | $-\frac{27}{2}{g_1}^2\,(M_F)_{ij}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$ | $-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$ | $-\frac{3}{2}{g_1}^2\,(M_F)_{ij}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $-\frac{9}{2}{g_2}^2\,(M_F)_{ij}$ | $-\frac{27}{2}{g_1}^2\,(M_F)_{ij}$ |

**Quantity:** $M_F$

# Terms 2 of 5

| Column | Representative term structure |
|---|---|
| T1 | $\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$ |
| T2 | $+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$ |
| T3 | $-6{g_1}^2\,(M_F)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$ | $+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$ | $-6{g_1}^2\,(M_F)_{ij}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$ | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}$ | $+\left(M_F\ (y_2)^\dagger\ y_2\right)_{ij}$ | $-6{g_1}^2\,(M_F)_{ij}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | $-6{g_1}^2\,(M_F)_{ij}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | $-6{g_1}^2\,(M_F)_{ij}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $M_F$

# Terms 3 of 5

| Column | Representative term structure |
|---|---|
| T1 | $\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}$ |
| T2 | $+\left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}$ |
| T3 | $+\left((y_2)^T\ (y_2)^*\ M_F\right)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}$ | $+\left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}$ | $+\left((y_2)^T\ (y_2)^*\ M_F\right)_{ij}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $M_F$

# Terms 4 of 5

| Column | Representative term structure |
|---|---|
| T1 | $\frac{1}{2}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ |
| T2 | $-12{g_2}^2\,(M_F)_{ij}$ |
| T3 | $\frac{1}{2}(\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij} + \left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij} + \left((y_2)^T\ (y_2)^*\ M_F\right)_{ij})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\frac{1}{2}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-12{g_2}^2\,(M_F)_{ij}$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | $\frac{1}{2}(\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij} + \left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij} + \left((y_2)^T\ (y_2)^*\ M_F\right)_{ij})$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\frac{1}{2}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | $-12{g_2}^2\,(M_F)_{ij}$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\frac{3}{4}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\frac{3}{4}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\frac{3}{4}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $\frac{3}{4}(\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij} + \left(M_F\ (y_2)^\dagger\ y_2\right)_{ij})$ | — | — |

**Quantity:** $M_F$

# Terms 5 of 5

| Column | Representative term structure |
|---|---|
| T1 | $-6{g_2}^2((\left((M_F)^T\right)_{ij} + (M_F)_{ij})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-6{g_2}^2((\left((M_F)^T\right)_{ij} + (M_F)_{ij})$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — |

$\beta^{(1)}_{m_1^2}$

**Quantity:** $m_1^2$

$$16\pi^2\,\mu\frac{dm_1^2}{d\mu} = \beta^{(1)}_{m_1^2}$$

# Terms 1 of 5

| Column | Representative term structure |
|---|---|
| T1 | $m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$ |
| T2 | $-8\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ |
| T3 | $-24{g_1}^2 m_1^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$ | $-8\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | $-24{g_1}^2 m_1^2$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$ | $-8\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | $-6{g_1}^2 m_1^2$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $m_1^2(4T_\nu^{(1)} + 4\lambda_{S_1}^{(1)})$ | $-8\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $-6\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $-6\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | $-6{g_1}^2 m_1^2$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | $-6{g_1}^2 m_1^2$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | $-24{g_1}^2 m_1^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | $-6{g_1}^2 m_1^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $-4\,\mathrm{Tr}\left(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\right)$ | $-6{g_1}^2 m_1^2$ |

**Quantity:** $m_1^2$

# Terms 2 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+6\lambda_{12}^{(1)} m_2^2$ |
| T2 | $m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$ |
| T3 | $-\frac{9{g_2}^2 m_1^2}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+6\lambda_{12}^{(1)} m_2^2$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+6\lambda_{12}^{(1)} m_2^2$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+6\lambda_{12}^{(1)} m_2^2$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+4\lambda_{12}^{(1)} m_2^2$ | $m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$ | $-\frac{9{g_2}^2 m_1^2}{2}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+4\lambda_{12}^{(1)} m_2^2$ | $m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$ | $-\frac{9{g_2}^2 m_1^2}{2}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+4\lambda_{12}^{(1)} m_2^2$ | $m_1^2(2T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$ | $-\frac{9{g_2}^2 m_1^2}{2}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+4\lambda_{12}^{(1)} m_2^2$ | — | $-\frac{9{g_2}^2 m_1^2}{2}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+4\lambda_{12}^{(1)} m_2^2$ | — | $-\frac{9{g_2}^2 m_1^2}{2}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4\lambda_{12}^{(1)} m_2^2$ | — | $-\frac{9{g_2}^2 m_1^2}{2}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+2\lambda_{12}^{(1)} m_2^2$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+2\lambda_{12}^{(1)} m_2^2$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+2\lambda_{12}^{(1)} m_2^2$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $m_1^2$

# Terms 3 of 5

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{27 {g_1}^2 m_1^2}{2}$ |
| T2 | $-8\,\mathrm{Tr}\left(M_F\ (M_F)^*\ (y_1)^T\ (y_1)^*\right)$ |
| T3 | $-8\,\mathrm{Tr}\left(M_F\ (y_1)^\dagger\ y_1\ (M_F)^*\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-\frac{27 {g_1}^2 m_1^2}{2}$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-\frac{3 {g_1}^2 m_1^2}{2}$ | $-8\,\mathrm{Tr}\left(M_F\ (M_F)^*\ (y_1)^T\ (y_1)^*\right)$ | $-8\,\mathrm{Tr}\left(M_F\ (y_1)^\dagger\ y_1\ (M_F)^*\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-\frac{3 {g_1}^2 m_1^2}{2}$ | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{27 {g_1}^2 m_1^2}{2}$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-\frac{3 {g_1}^2 m_1^2}{2}$ | $-12\,\mathrm{Tr}\left(M_F\ (M_F)^*\ (y_1)^T\ (y_1)^*\right)$ | $-12\,\mathrm{Tr}\left(M_F\ (y_1)^\dagger\ y_1\ (M_F)^*\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{3 {g_1}^2 m_1^2}{2}$ | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $m_1^2$

# Terms 4 of 5

| Column | Representative term structure |
|---|---|
| T1 | $m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$ |
| T2 | $m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$ |
| T3 | $-12{g_2}^2 m_1^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$ | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$ | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$ | $-12{g_2}^2 m_1^2$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$ | $-12{g_2}^2 m_1^2$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$ | $-12{g_2}^2 m_1^2$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$ | $-12{g_2}^2 m_1^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$ | $-12{g_2}^2 m_1^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$ | $-12{g_2}^2 m_1^2$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$ | $-12{g_2}^2 m_1^2$ |

**Quantity:** $m_1^2$

# Terms 5 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$ |

$\beta^{(1)}_{m_2^2}$

**Quantity:** $m_2^2$

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu} = \beta^{(1)}_{m_2^2}$$

## Terms 1 of 5

| Column | Representative term structure |
|---|---|
| T1 | $m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$ |
| T2 | $-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ |
| T3 | $-12{g_2}^2 m_2^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$ | $-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | $-12{g_2}^2 m_2^2$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$ | $-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | $-12{g_2}^2 m_2^2$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$ | $-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | $-12{g_2}^2 m_2^2$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $-6\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $-6\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $-8\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $-8\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $-8\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$ | $-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | $-12{g_2}^2 m_2^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$ | $-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | $-12{g_2}^2 m_2^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$ | $-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | $-12{g_2}^2 m_2^2$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$ | $-4\,\mathrm{Tr}\left(M_F\ (y_2)^\dagger\ y_2\ (M_F)^\dagger\right)$ | $-12{g_2}^2 m_2^2$ |

**Quantity:** $m_2^2$

# Terms 2 of 5

| Column | Representative term structure |
|---|---|
| T1 | $-6{g_1}^2 m_2^2$ |
| T2 | $+2\lambda_{12}^{(1)} m_1^2$ |
| T3 | $m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-6{g_1}^2 m_2^2$ | $+2\lambda_{12}^{(1)} m_1^2$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $+2\lambda_{12}^{(1)} m_1^2$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-6{g_1}^2 m_2^2$ | $+2\lambda_{12}^{(1)} m_1^2$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $+4\lambda_{12}^{(1)} m_1^2$ | $m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $+4\lambda_{12}^{(1)} m_1^2$ | $m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $+4\lambda_{12}^{(1)} m_1^2$ | $m_2^2(2T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $+4\lambda_{12}^{(1)} m_1^2$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $+4\lambda_{12}^{(1)} m_1^2$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $+4\lambda_{12}^{(1)} m_1^2$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $+6\lambda_{12}^{(1)} m_1^2$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $-6{g_1}^2 m_2^2$ | $+6\lambda_{12}^{(1)} m_1^2$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $-24{g_1}^2 m_2^2$ | $+6\lambda_{12}^{(1)} m_1^2$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-6{g_1}^2 m_2^2$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-6{g_1}^2 m_2^2$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $-24{g_1}^2 m_2^2$ | — | — |

**Quantity:** $m_2^2$

# Terms 3 of 5

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{9{g_2}^2 m_2^2}{2}$ |
| T2 | $-\frac{3{g_1}^2 m_2^2}{2}$ |
| T3 | $-8\,\mathrm{Tr}\left(M_F\ (M_F)^*\ (y_2)^T\ (y_2)^*\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-\frac{9{g_2}^2 m_2^2}{2}$ | $-\frac{3{g_1}^2 m_2^2}{2}$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-\frac{9{g_2}^2 m_2^2}{2}$ | $-\frac{3{g_1}^2 m_2^2}{2}$ | $-8\,\mathrm{Tr}\left(M_F\ (M_F)^*\ (y_2)^T\ (y_2)^*\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-\frac{9{g_2}^2 m_2^2}{2}$ | $-\frac{27{g_1}^2 m_2^2}{2}$ | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-\frac{9{g_2}^2 m_2^2}{2}$ | $-\frac{3{g_1}^2 m_2^2}{2}$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-\frac{9{g_2}^2 m_2^2}{2}$ | $-\frac{3{g_1}^2 m_2^2}{2}$ | $-12\,\mathrm{Tr}\left(M_F\ (M_F)^*\ (y_2)^T\ (y_2)^*\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-\frac{9{g_2}^2 m_2^2}{2}$ | $-\frac{27{g_1}^2 m_2^2}{2}$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $m_2^2$

# Terms 4 of 5

| Column | Representative term structure |
|---|---|
| T1 | $-8\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^*\Big)$ |
| T2 | $m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$ |
| T3 | $m_2^2(4T_\nu^{(2)} + 4\lambda_{S_2}^{(1)})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-8\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^*\Big)$ | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-12\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^*\Big)$ | $m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | $m_2^2(4T_\nu^{(2)} + 4\lambda_{S_2}^{(1)})$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | $m_2^2(4T_\nu^{(2)} + 4\lambda_{S_2}^{(1)})$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | $m_2^2(4T_\nu^{(2)} + 4\lambda_{S_2}^{(1)})$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $m_2^2$

# Terms 5 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$ |

$\beta^{(1)}_{\lambda_1}$

**Quantity:** $\lambda_1$

$$16\pi^2\,\mu\frac{d\lambda_1}{d\mu} = \beta^{(1)}_{\lambda_1}$$

# Terms 1 of 7

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ |
| T3 | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $\lambda_1(12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-12\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\right)$ | $-4\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |

**Quantity:** $\lambda_1$

# Terms 2 of 7

| Column | Representative term structure |
|---|---|
| T1 | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ |
| T2 | $+4\lambda_5\ (\lambda_5)^*$ |
| T3 | $+\frac{9{g_2}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $-12\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ | $+\frac{9{g_2}^4}{4}$ |

**Quantity:** $\lambda_1$

# Terms 3 of 7

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ |
| T2 | $-9{g_2}^2\lambda_1$ |
| T3 | $+\frac{3{g_1}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$ | $-9{g_2}^2\lambda_1$ | $+\frac{3{g_1}^4}{4}$ |

**Quantity:** $\lambda_1$

# Terms 4 of 7

| Column | Representative term structure |
|---|---|
| T1 | $+12{\lambda_1}^2$ |
| T2 | $+2{\lambda_{HS_1}^{(1)}}^2$ |
| T3 | $+6{\lambda_{HS_2}^{(1)}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+12{\lambda_1}^2$ | $+2{\lambda_{HS_1}^{(1)}}^2$ | $+6{\lambda_{HS_2}^{(1)}}^2$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+12{\lambda_1}^2$ | $+2{\lambda_{HS_1}^{(1)}}^2$ | $+6{\lambda_{HS_2}^{(1)}}^2$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+12{\lambda_1}^2$ | $+2{\lambda_{HS_1}^{(1)}}^2$ | $+6{\lambda_{HS_2}^{(1)}}^2$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+12{\lambda_1}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+12{\lambda_1}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+12{\lambda_1}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+12{\lambda_1}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+12{\lambda_1}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+12{\lambda_1}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+12{\lambda_1}^2$ | $+6{\lambda_{HS_1}^{(1)}}^2$ | $+2{\lambda_{HS_2}^{(1)}}^2$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+12{\lambda_1}^2$ | $+6{\lambda_{HS_1}^{(1)}}^2$ | $+2{\lambda_{HS_2}^{(1)}}^2$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+12{\lambda_1}^2$ | $+6{\lambda_{HS_1}^{(1)}}^2$ | $+2{\lambda_{HS_2}^{(1)}}^2$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+12{\lambda_1}^2$ | $+6{\lambda_{HS_1}^{(1)}}^2$ | $+6{\lambda_{HS_2}^{(1)}}^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+12{\lambda_1}^2$ | $+6{\lambda_{HS_1}^{(1)}}^2$ | $+6{\lambda_{HS_2}^{(1)}}^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+12{\lambda_1}^2$ | $+6{\lambda_{HS_1}^{(1)}}^2$ | $+6{\lambda_{HS_2}^{(1)}}^2$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+12{\lambda_1}^2$ | $+6{\lambda_{HS_1}^{(1)}}^2$ | $+6{\lambda_{HS_2}^{(1)}}^2$ |

**Quantity:** $\lambda_1$

# Terms 5 of 7

| Column | Representative term structure |
|---|---|
| T1 | $+3{\lambda_{HS_2}^{(A)}}^2$ |
| T2 | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ |
| T3 | $+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+3{\lambda_{HS_2}^{(A)}}^2$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+3{\lambda_{HS_2}^{(A)}}^2$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+3{\lambda_{HS_2}^{(A)}}^2$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ | $+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ | $+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ | $+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ | $+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ | $+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $+\frac{{\lambda_{HS_1}^{(A)}}^2}{4}$ | $+\frac{{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+3{\lambda_{HS_2}^{(A)}}^2$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+3{\lambda_{HS_2}^{(A)}}^2$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+3{\lambda_{HS_2}^{(A)}}^2$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+3{\lambda_{HS_2}^{(A)}}^2$ | — | — |

**Quantity:** $\lambda_1$

# Terms 6 of 7

| Column | Representative term structure |
|---|---|
| T1 | $+8\lambda_7\,(\lambda_7)^*$ |
| T2 | $+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ |
| T3 | $+8\lambda_8\,(\lambda_8)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+8\lambda_7\,(\lambda_7)^*$ | $+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ | $+8\lambda_8\,(\lambda_8)^*$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+8\lambda_7\,(\lambda_7)^*$ | $+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ | $+8\lambda_8\,(\lambda_8)^*$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_1$

# Terms 7 of 7

| Column | Representative term structure |
|---|---|
| T1 | $+3\lambda_{HS_1}^{(A)\ 2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+3\lambda_{HS_1}^{(A)\ 2}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+3\lambda_{HS_1}^{(A)\ 2}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+3\lambda_{HS_1}^{(A)\ 2}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+3\lambda_{HS_1}^{(A)\ 2}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+3\lambda_{HS_1}^{(A)\ 2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+3\lambda_{HS_1}^{(A)\ 2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+3\lambda_{HS_1}^{(A)\ 2}$ |

$\beta^{(1)}_{\lambda^{(1)}_{S_1}}$

**Quantity:** $\lambda^{(1)}_{S_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_1}}$$

## Terms 1 of 9

| Column | Representative term structure |
|---|---|
| T1 | $8\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$ |
| T2 | $-8\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ |
| T3 | $+192{g_1}^4$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $8\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$ | $-8\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | $+192{g_1}^4$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $8\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$ | $-8\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | $+12{g_1}^4$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $8\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$ | $-8\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $4\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$ | $-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $4\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$ | $-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $4\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$ | $-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $6\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$ | $-5\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $6\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$ | $-5\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $6\lambda^{(1)}_{S_1}T^{(1)}_{\nu}$ | $-5\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | $+12{g_1}^4$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | $+12{g_1}^4$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | $+192{g_1}^4$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | $+12{g_1}^4$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ | $+12{g_1}^4$ |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 2 of 9

| Column | Representative term structure |
|---|---|
| T1 | $-48{g_1}^2\lambda_{S_1}^{(1)}$ |
| T2 | $+6{\lambda_{12}^{(1)}}^2$ |
| T3 | $+4{\lambda_{HS_1}^{(1)}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-48{g_1}^2\lambda_{S_1}^{(1)}$ | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-12{g_1}^2\lambda_{S_1}^{(1)}$ | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $+4{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $+4{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $+4{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $+4{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $+4{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $+4{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $+2{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $+2{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $+2{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_1}^{(1)}}^2$ |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 3 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+10{\lambda_{S_1}^{(1)}}^2$ |
| T2 | $+\frac{9{g_2}^4}{4}$ |
| T3 | $+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_1}^{(1)}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+10{\lambda_{S_1}^{(1)}}^2$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+10{\lambda_{S_1}^{(1)}}^2$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+10{\lambda_{S_1}^{(1)}}^2$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+12{\lambda_{S_1}^{(1)}}^2$ | $+\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_1}^{(1)}\right)$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+12{\lambda_{S_1}^{(1)}}^2$ | $+\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+12{\lambda_{S_1}^{(1)}}^2$ | $+\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+12{\lambda_{S_1}^{(1)}}^2$ | $+\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{27{g_2}^2}{2}-27\lambda_{S_1}^{(1)}\right)$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+12{\lambda_{S_1}^{(1)}}^2$ | $+\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+12{\lambda_{S_1}^{(1)}}^2$ | $+\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_{S_1}^{(1)}\right)$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+14{\lambda_{S_1}^{(1)}}^2$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+14{\lambda_{S_1}^{(1)}}^2$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+14{\lambda_{S_1}^{(1)}}^2$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+14{\lambda_{S_1}^{(1)}}^2$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+14{\lambda_{S_1}^{(1)}}^2$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+14{\lambda_{S_1}^{(1)}}^2$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+14{\lambda_{S_1}^{(1)}}^2$ | — | — |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 4 of 9

| Column | Representative term structure |
|---|---|
| T1 | $-9{g_2}^2\lambda_{S_1}^{(1)}$ |
| T2 | $+\frac{243{g_1}^4}{4}$ |
| T3 | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-9{g_2}^2\lambda_{S_1}^{(1)}$ | $+\frac{243{g_1}^4}{4}$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-9{g_2}^2\lambda_{S_1}^{(1)}$ | $+\frac{3{g_1}^4}{4}$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-9{g_2}^2\lambda_{S_1}^{(1)}$ | $+\frac{3{g_1}^4}{4}$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-9{g_2}^2\lambda_{S_1}^{(1)}$ | $+\frac{243{g_1}^4}{4}$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-9{g_2}^2\lambda_{S_1}^{(1)}$ | $+\frac{3{g_1}^4}{4}$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-9{g_2}^2\lambda_{S_1}^{(1)}$ | $+\frac{3{g_1}^4}{4}$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 5 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$ |
| T2 | $+\lambda_{11}(2\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right) + 24\,(\lambda_{11})^*)$ |
| T3 | $+2\,(\lambda_{11})^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$ | $+\lambda_{11}(2\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right) + 24\,(\lambda_{11})^*)$ | $+2\,(\lambda_{11})^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$ | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$ | $+\lambda_{11}(3\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right) + 24\,(\lambda_{11})^*)$ | $+3\,(\lambda_{11})^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\frac{\lambda_{HS_1}^{(A)\ 2}}{4}$ | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 6 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+8\lambda_7\,(\lambda_7)^*$ |
| T2 | $+8\lambda_9\,(\lambda_9)^*$ |
| T3 | $\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+8\lambda_7\,(\lambda_7)^*$ | $+8\lambda_9\,(\lambda_9)^*$ | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+8\lambda_7\,(\lambda_7)^*$ | $+8\lambda_9\,(\lambda_9)^*$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | $\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | $\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | $\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | $\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | $\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | $\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | $\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$ |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 7 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_{S_1}^{(A)} T_\nu^{(1)}$ |
| T2 | $+21{g_2}^4$ |
| T3 | $+{g_1}^2 \left(12{g_2}^2 - 12\lambda_{S_1}^{(1)} - 6\lambda_{S_1}^{(A)}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+2\lambda_{S_1}^{(A)} T_\nu^{(1)}$ | $+21{g_2}^4$ | $+{g_1}^2 \left(12{g_2}^2 - 12\lambda_{S_1}^{(1)} - 6\lambda_{S_1}^{(A)}\right)$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+2\lambda_{S_1}^{(A)} T_\nu^{(1)}$ | $+21{g_2}^4$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+2\lambda_{S_1}^{(A)} T_\nu^{(1)}$ | $+21{g_2}^4$ | $+{g_1}^2 \left(12{g_2}^2 - 12\lambda_{S_1}^{(1)} - 6\lambda_{S_1}^{(A)}\right)$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+2\lambda_{S_1}^{(A)} T_\nu^{(1)}$ | $+21{g_2}^4$ | $+{g_1}^2 \left(48{g_2}^2 - 48\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)}\right)$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+2\lambda_{S_1}^{(A)} T_\nu^{(1)}$ | $+21{g_2}^4$ | $+{g_1}^2 \left(12{g_2}^2 - 12\lambda_{S_1}^{(1)} - 6\lambda_{S_1}^{(A)}\right)$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+2\lambda_{S_1}^{(A)} T_\nu^{(1)}$ | $+21{g_2}^4$ | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+2\lambda_{S_1}^{(A)} T_\nu^{(1)}$ | $+21{g_2}^4$ | $+{g_1}^2 \left(12{g_2}^2 - 12\lambda_{S_1}^{(1)} - 6\lambda_{S_1}^{(A)}\right)$ |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 8 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+{g_2}^2(-24\lambda_{S_1}^{(1)} - 12\lambda_{S_1}^{(A)})$ |
| T2 | $+\frac{3{\lambda_{HS_1}^{(A)}}^2}{2}$ |
| T3 | $+26{\lambda_{S_1}^{(A)}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+{g_2}^2(-24\lambda_{S_1}^{(1)} - 12\lambda_{S_1}^{(A)})$ | $+\frac{3{\lambda_{HS_1}^{(A)}}^2}{2}$ | $+26{\lambda_{S_1}^{(A)}}^2$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+{g_2}^2(-24\lambda_{S_1}^{(1)} - 12\lambda_{S_1}^{(A)})$ | $+\frac{3{\lambda_{HS_1}^{(A)}}^2}{2}$ | $+26{\lambda_{S_1}^{(A)}}^2$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+{g_2}^2(-24\lambda_{S_1}^{(1)} - 12\lambda_{S_1}^{(A)})$ | $+\frac{3{\lambda_{HS_1}^{(A)}}^2}{2}$ | $+26{\lambda_{S_1}^{(A)}}^2$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+{g_2}^2(-24\lambda_{S_1}^{(1)} - 12\lambda_{S_1}^{(A)})$ | $+\frac{3{\lambda_{HS_1}^{(A)}}^2}{2}$ | $+26{\lambda_{S_1}^{(A)}}^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+{g_2}^2(-24\lambda_{S_1}^{(1)} - 12\lambda_{S_1}^{(A)})$ | $+\frac{3{\lambda_{HS_1}^{(A)}}^2}{2}$ | $+26{\lambda_{S_1}^{(A)}}^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+{g_2}^2(-24\lambda_{S_1}^{(1)} - 12\lambda_{S_1}^{(A)})$ | $+\frac{3{\lambda_{HS_1}^{(A)}}^2}{2}$ | $+26{\lambda_{S_1}^{(A)}}^2$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+{g_2}^2(-24\lambda_{S_1}^{(1)} - 12\lambda_{S_1}^{(A)})$ | $+\frac{3{\lambda_{HS_1}^{(A)}}^2}{2}$ | $+26{\lambda_{S_1}^{(A)}}^2$ |

**Quantity:** $\lambda_{S_1}^{(1)}$

# Terms 9 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ |
| T2 | $+6\lambda_{12}^{(A)\,2}$ |
| T3 | $+6\lambda_{12}^{(\times)\,2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ | $+6\lambda_{12}^{(A)\,2}$ | $+6\lambda_{12}^{(\times)\,2}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ | $+6\lambda_{12}^{(A)\,2}$ | $+6\lambda_{12}^{(\times)\,2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ | $+6\lambda_{12}^{(A)\,2}$ | $+6\lambda_{12}^{(\times)\,2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ | $+6\lambda_{12}^{(A)\,2}$ | $+6\lambda_{12}^{(\times)\,2}$ |

$\beta^{(1)}_{\lambda^{(1)}_{S_2}}$

**Quantity:** $\lambda^{(1)}_{S_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_2}}$$

## Terms 1 of 9

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 14\lambda^{(A)}_{S_2})$ |
| T2 | $+2\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ |
| T3 | $-3\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 14\lambda^{(A)}_{S_2})$ | $+2\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-3\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 14\lambda^{(A)}_{S_2})$ | $+2\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-3\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 14\lambda^{(A)}_{S_2})$ | $+2\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-3\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | $-4\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | $-4\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | $-4\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | $-5\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | $-5\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | $-5\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | $-8\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | $-8\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | $-8\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 14\lambda^{(A)}_{S_2})$ | $+2\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-3\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 14\lambda^{(A)}_{S_2})$ | $+2\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-3\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 14\lambda^{(A)}_{S_2})$ | $+2\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-3\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $\lambda^{(1)}_{S_2}(4T^{(2)}_{\nu} + 14\lambda^{(A)}_{S_2})$ | $+2\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-3\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |

**Quantity:** $\lambda_{S_2}^{(1)}$

# Terms 2 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+21{g_2}^4$ |
| T2 | $+{g_1}^2\left(12{g_2}^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$ |
| T3 | $+{g_2}^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+21{g_2}^4$ | $+{g_1}^2\left(12{g_2}^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$ | $+{g_2}^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+21{g_2}^4$ | — | $+{g_2}^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+21{g_2}^4$ | $+{g_1}^2\left(12{g_2}^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$ | $+{g_2}^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+21{g_2}^4$ | $+{g_1}^2\left(12{g_2}^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$ | $+{g_2}^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+21{g_2}^4$ | — | $+{g_2}^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+21{g_2}^4$ | $+{g_1}^2\left(12{g_2}^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$ | $+{g_2}^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+21{g_2}^4$ | $+{g_1}^2\left(48{g_2}^2-48\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)}\right)$ | $+{g_2}^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$ |

**Quantity:** $\lambda_{S_2}^{(1)}$

# Terms 3 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+12{g_1}^4$ |
| T2 | $+2{\lambda_{12}^{(1)}}^2$ |
| T3 | $+4{\lambda_{HS_2}^{(1)}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+12{g_1}^4$ | $+2{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $+2{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+12{g_1}^4$ | $+2{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $+4{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $+4{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $+4{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $+4{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $+4{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $+4{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+12{g_1}^4$ | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+192{g_1}^4$ | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+12{g_1}^4$ | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+12{g_1}^4$ | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+192{g_1}^4$ | $+6{\lambda_{12}^{(1)}}^2$ | $+4{\lambda_{HS_2}^{(1)}}^2$ |

**Quantity:** $\lambda_{S_2}^{(1)}$

# Terms 4 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3\lambda_{HS_2}^{(A)\ 2}}{2}$ |
| T2 | $+14\lambda_{S_2}^{(1)\,2}$ |
| T3 | $+26\lambda_{S_2}^{(A)\,2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{3\lambda_{HS_2}^{(A)\ 2}}{2}$ | $+14\lambda_{S_2}^{(1)\,2}$ | $+26\lambda_{S_2}^{(A)\,2}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3\lambda_{HS_2}^{(A)\ 2}}{2}$ | $+14\lambda_{S_2}^{(1)\,2}$ | $+26\lambda_{S_2}^{(A)\,2}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3\lambda_{HS_2}^{(A)\ 2}}{2}$ | $+14\lambda_{S_2}^{(1)\,2}$ | $+26\lambda_{S_2}^{(A)\,2}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $+12\lambda_{S_2}^{(1)\,2}$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $+12\lambda_{S_2}^{(1)\,2}$ | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $+12\lambda_{S_2}^{(1)\,2}$ | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $+12\lambda_{S_2}^{(1)\,2}$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $+12\lambda_{S_2}^{(1)\,2}$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $+12\lambda_{S_2}^{(1)\,2}$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $+10\lambda_{S_2}^{(1)\,2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $+10\lambda_{S_2}^{(1)\,2}$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $+10\lambda_{S_2}^{(1)\,2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{3\lambda_{HS_2}^{(A)\ 2}}{2}$ | $+14\lambda_{S_2}^{(1)\,2}$ | $+26\lambda_{S_2}^{(A)\,2}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3\lambda_{HS_2}^{(A)\ 2}}{2}$ | $+14\lambda_{S_2}^{(1)\,2}$ | $+26\lambda_{S_2}^{(A)\,2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3\lambda_{HS_2}^{(A)\ 2}}{2}$ | $+14\lambda_{S_2}^{(1)\,2}$ | $+26\lambda_{S_2}^{(A)\,2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\frac{3\lambda_{HS_2}^{(A)\ 2}}{2}$ | $+14\lambda_{S_2}^{(1)\,2}$ | $+26\lambda_{S_2}^{(A)\,2}$ |

**Quantity:** $\lambda_{S_2}^{(1)}$

# Terms 5 of 9

| Column | Representative term structure |
|---|---|
| T1 | $4\lambda_{S_2}^{(1)} T_\nu^{(2)}$ |
| T2 | $+\frac{9{g_2}^4}{4}$ |
| T3 | $+{g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_{S_2}^{(1)}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $4\lambda_{S_2}^{(1)} T_\nu^{(2)}$ | $+\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_{S_2}^{(1)}\right)$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+4\lambda_{S_2}^{(1)} T_\nu^{(2)}$ | $+\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_{S_2}^{(1)}\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $4\lambda_{S_2}^{(1)} T_\nu^{(2)}$ | $+\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{27{g_2}^2}{2} - 27\lambda_{S_2}^{(1)}\right)$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $6\lambda_{S_2}^{(1)} T_\nu^{(2)}$ | $+\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_{S_2}^{(1)}\right)$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+6\lambda_{S_2}^{(1)} T_\nu^{(2)}$ | $+\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_{S_2}^{(1)}\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $6\lambda_{S_2}^{(1)} T_\nu^{(2)}$ | $+\frac{9{g_2}^4}{4}$ | $+{g_1}^2\left(\frac{27{g_2}^2}{2} - 27\lambda_{S_2}^{(1)}\right)$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $8\lambda_{S_2}^{(1)} T_\nu^{(2)}$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $8\lambda_{S_2}^{(1)} T_\nu^{(2)}$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $8\lambda_{S_2}^{(1)} T_\nu^{(2)}$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{S_2}^{(1)}$

# Terms 6 of 9

| Column | Representative term structure |
|---|---|
| T1 | $-9{g_2}^2\lambda_{S_2}^{(1)}$ |
| T2 | $+\frac{3{g_1}^4}{4}$ |
| T3 | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-9{g_2}^2\lambda_{S_2}^{(1)}$ | $+\frac{3{g_1}^4}{4}$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-9{g_2}^2\lambda_{S_2}^{(1)}$ | $+\frac{3{g_1}^4}{4}$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-9{g_2}^2\lambda_{S_2}^{(1)}$ | $+\frac{243{g_1}^4}{4}$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-9{g_2}^2\lambda_{S_2}^{(1)}$ | $+\frac{3{g_1}^4}{4}$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-9{g_2}^2\lambda_{S_2}^{(1)}$ | $+\frac{3{g_1}^4}{4}$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-9{g_2}^2\lambda_{S_2}^{(1)}$ | $+\frac{243{g_1}^4}{4}$ | $+\frac{{\lambda_{12}^{(A)}}^2}{4}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{S_2}^{(1)}$

# Terms 7 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ |
| T2 | $\lambda_{10}(2\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right) + 24\,(\lambda_{10})^*)$ |
| T3 | $+2\,(\lambda_{10})^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ | $\lambda_{10}(2\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right) + 24\,(\lambda_{10})^*)$ | $+2\,(\lambda_{10})^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ | $\lambda_{10}(3\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right) + 24\,(\lambda_{10})^*)$ | $+3\,(\lambda_{10})^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\frac{\lambda_{HS_2}^{(A)\ 2}}{4}$ | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{S_2}^{(1)}$

# Terms 8 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+8\lambda_8 \left(\lambda_8\right)^*$ |
| T2 | $+8\lambda_9 \left(\lambda_9\right)^*$ |
| T3 | $-12{g_1}^2\lambda_{S_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+8\lambda_8 \left(\lambda_8\right)^*$ | $+8\lambda_9 \left(\lambda_9\right)^*$ | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+8\lambda_8 \left(\lambda_8\right)^*$ | $+8\lambda_9 \left(\lambda_9\right)^*$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | $-12{g_1}^2\lambda_{S_2}^{(1)}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | $-48{g_1}^2\lambda_{S_2}^{(1)}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda^{(1)}_{S_2}$

# Terms 9 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+8\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$ |
| T2 | $+6{\lambda^{(A)}_{12}}^2$ |
| T3 | $+6{\lambda^{(\times)}_{12}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+8\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$ | $+6{\lambda^{(A)}_{12}}^2$ | $+6{\lambda^{(\times)}_{12}}^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+8\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$ | $+6{\lambda^{(A)}_{12}}^2$ | $+6{\lambda^{(\times)}_{12}}^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+8\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$ | $+6{\lambda^{(A)}_{12}}^2$ | $+6{\lambda^{(\times)}_{12}}^2$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+8\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$ | $+6{\lambda^{(A)}_{12}}^2$ | $+6{\lambda^{(\times)}_{12}}^2$ |

$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$

**Quantity:** $\lambda^{(1)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_1}}$$

## Terms 1 of 9

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{HS_1}(4T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3 | $+12\lambda_5\ (\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda^{(1)}_{HS_1}(4T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+12\lambda_5\ (\lambda_5)^*$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(1)}_{HS_1}(4T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+12\lambda_5\ (\lambda_5)^*$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(1)}_{HS_1}(4T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+12\lambda_5\ (\lambda_5)^*$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\ (\lambda_5)^*$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\ (\lambda_5)^*$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\ (\lambda_5)^*$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\ (\lambda_5)^*$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\ (\lambda_5)^*$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $-3\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\ (\lambda_5)^*$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $-2\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\ (\lambda_5)^*$ |

**Quantity:** $\lambda_{HS_1}^{(1)}$

# Terms 2 of 9

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}$ |
| T2 | $+12{g_1}^4$ |
| T3 | $-\frac{51{g_1}^2\lambda_{HS_1}^{(1)}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}$ | $+12{g_1}^4$ | $-\frac{51{g_1}^2\lambda_{HS_1}^{(1)}}{2}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}$ | $+3{g_1}^4$ | $-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}$ | — | $-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$ | $+3{g_1}^4$ | $-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$ | — | $-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$ | $+3{g_1}^4$ | $-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$ | $+12{g_1}^4$ | $-\frac{51{g_1}^2\lambda_{HS_1}^{(1)}}{2}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$ | $+3{g_1}^4$ | $-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$ | — | $-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $-\frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2}$ | $+3{g_1}^4$ | $-\frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2}$ |

**Quantity:** $\lambda^{(1)}_{HS_1}$

# Terms 3 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+6\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ |
| T2 | $+6\lambda_1\lambda^{(1)}_{HS_1}$ |
| T3 | $+4{\lambda^{(1)}_{HS_1}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+6\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+6\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+6\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+2\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+2\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+2\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$ | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $+6\lambda_1\lambda^{(1)}_{HS_1}$ | $+4{\lambda^{(1)}_{HS_1}}^2$ |

**Quantity:** $\lambda^{(1)}_{HS_1}$

# Terms 4 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ |
| T2 | $\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T3 | $+\frac{9{g_2}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+4\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | $\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+\frac{9{g_2}^4}{4}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | $\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+\frac{9{g_2}^4}{4}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | $\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+\frac{9{g_2}^4}{4}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | $+\frac{9{g_2}^4}{4}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | $+\frac{9{g_2}^4}{4}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+6\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | $+\frac{9{g_2}^4}{4}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+8\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$ | — | — |

**Quantity:** $\lambda^{(1)}_{HS_1}$

# Terms 5 of 9

| Column | Representative term structure |
|---|---|
| T1 | $-9{g_2}^2\lambda^{(1)}_{HS_1}$ |
| T2 | $+\frac{27{g_1}^4}{4}$ |
| T3 | $-15{g_1}^2\lambda^{(1)}_{HS_1}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-9{g_2}^2\lambda^{(1)}_{HS_1}$ | $+\frac{27{g_1}^4}{4}$ | $-15{g_1}^2\lambda^{(1)}_{HS_1}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-9{g_2}^2\lambda^{(1)}_{HS_1}$ | $+\frac{3{g_1}^4}{4}$ | $-3{g_1}^2\lambda^{(1)}_{HS_1}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-9{g_2}^2\lambda^{(1)}_{HS_1}$ | $+\frac{3{g_1}^4}{4}$ | $-3{g_1}^2\lambda^{(1)}_{HS_1}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-9{g_2}^2\lambda^{(1)}_{HS_1}$ | $+\frac{27{g_1}^4}{4}$ | $-15{g_1}^2\lambda^{(1)}_{HS_1}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-9{g_2}^2\lambda^{(1)}_{HS_1}$ | $+\frac{3{g_1}^4}{4}$ | $-3{g_1}^2\lambda^{(1)}_{HS_1}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-9{g_2}^2\lambda^{(1)}_{HS_1}$ | $+\frac{3{g_1}^4}{4}$ | $-3{g_1}^2\lambda^{(1)}_{HS_1}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda^{(1)}_{HS_1}$

# Terms 6 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3\lambda^{(A)}_{HS_1}{}^2}{4}$ |
| T2 | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\frac{1}{2}\,\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*+3\,(\lambda_{11})^*\right)$ |
| T3 | $+\frac{1}{2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+\frac{3\lambda^{(A)}_{HS_1}{}^2}{4}$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\frac{3\lambda^{(A)}_{HS_1}{}^2}{4}$ | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\frac{1}{2}\,\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*+3\,(\lambda_{11})^*\right)$ | $+\frac{1}{2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+\frac{3\lambda^{(A)}_{HS_1}{}^2}{4}$ | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{3\lambda^{(A)}_{HS_1}{}^2}{4}$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{3\lambda^{(A)}_{HS_1}{}^2}{4}$ | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\frac{3}{4}\,\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*+3\,(\lambda_{11})^*\right)$ | $+\frac{3}{4}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\frac{3\lambda^{(A)}_{HS_1}{}^2}{4}$ | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda^{(1)}_{HS_1}$

# Terms 7 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+24\lambda_7\,(\lambda_7)^*$ |
| T2 | $+3\lambda_{11}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ |
| T3 | $\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+24\lambda_7\,(\lambda_7)^*$ | $+3\lambda_{11}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | $\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+24\lambda_7\,(\lambda_7)^*$ | $+3\lambda_{11}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ | $\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | $\lambda^{(1)}_{HS_1}(3T^{(1)}_\nu + 6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda^{(1)}_{HS_1}$

# Terms 8 of 9

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda^{(A)}_{S_1})$ |
| T2 | $+6{g_2}^4$ |
| T3 | $+6{\lambda^{(A)}_{HS_1}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda^{(A)}_{S_1})$ | $+6{g_2}^4$ | $+6{\lambda^{(A)}_{HS_1}}^2$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda^{(A)}_{S_1})$ | $+6{g_2}^4$ | $+6{\lambda^{(A)}_{HS_1}}^2$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda^{(A)}_{S_1})$ | $+6{g_2}^4$ | $+6{\lambda^{(A)}_{HS_1}}^2$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda^{(A)}_{S_1})$ | $+6{g_2}^4$ | $+6{\lambda^{(A)}_{HS_1}}^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda^{(A)}_{S_1})$ | $+6{g_2}^4$ | $+6{\lambda^{(A)}_{HS_1}}^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda^{(A)}_{S_1})$ | $+6{g_2}^4$ | $+6{\lambda^{(A)}_{HS_1}}^2$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $\lambda^{(1)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\ (Y_u)^\dagger\big) + 4\lambda^{(A)}_{S_1})$ | $+6{g_2}^4$ | $+6{\lambda^{(A)}_{HS_1}}^2$ |

**Quantity:** $\lambda^{(1)}_{HS_1}$

# Terms 9 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$ |

$\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$

**Quantity:** $\lambda^{(1)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_2}}$$

## Terms 1 of 9

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$ |
| T2 | $-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3 | $+4\lambda_5\,(\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$ | $-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\,(\lambda_5)^*$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$ | $-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\,(\lambda_5)^*$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$ | $-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\,(\lambda_5)^*$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\,(\lambda_5)^*$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\,(\lambda_5)^*$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\,(\lambda_5)^*$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $-3\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\,(\lambda_5)^*$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $-3\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\,(\lambda_5)^*$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $-3\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+6\lambda_5\,(\lambda_5)^*$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+12\lambda_5\,(\lambda_5)^*$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+12\lambda_5\,(\lambda_5)^*$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+12\lambda_5\,(\lambda_5)^*$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$ | $-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\,(\lambda_5)^*$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$ | $-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\,(\lambda_5)^*$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$ | $-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\,(\lambda_5)^*$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{S_2})$ | $-2\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+4\lambda_5\,(\lambda_5)^*$ |

**Quantity:** $\lambda^{(1)}_{HS_2}$

# Terms 2 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+6{g_2}^4$ |
| T2 | $-\frac{33{g_2}^2\lambda^{(1)}_{HS_2}}{2}$ |
| T3 | $+3{g_1}^4$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+6{g_2}^4$ | $-\frac{33{g_2}^2\lambda^{(1)}_{HS_2}}{2}$ | $+3{g_1}^4$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+6{g_2}^4$ | $-\frac{33{g_2}^2\lambda^{(1)}_{HS_2}}{2}$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+6{g_2}^4$ | $-\frac{33{g_2}^2\lambda^{(1)}_{HS_2}}{2}$ | $+3{g_1}^4$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $-\frac{9{g_2}^2\lambda^{(1)}_{HS_2}}{2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{9{g_2}^2\lambda^{(1)}_{HS_2}}{2}$ | $+3{g_1}^4$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $-\frac{9{g_2}^2\lambda^{(1)}_{HS_2}}{2}$ | $+12{g_1}^4$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+6{g_2}^4$ | $-\frac{33{g_2}^2\lambda^{(1)}_{HS_2}}{2}$ | $+3{g_1}^4$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+6{g_2}^4$ | $-\frac{33{g_2}^2\lambda^{(1)}_{HS_2}}{2}$ | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+6{g_2}^4$ | $-\frac{33{g_2}^2\lambda^{(1)}_{HS_2}}{2}$ | $+3{g_1}^4$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+6{g_2}^4$ | $-\frac{33{g_2}^2\lambda^{(1)}_{HS_2}}{2}$ | $+12{g_1}^4$ |

**Quantity:** $\lambda^{(1)}_{HS_2}$

# Terms 3 of 9

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{15{g_1}^2\lambda^{(1)}_{HS_2}}{2}$ |
| T2 | $+2\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ |
| T3 | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{15{g_1}^2\lambda^{(1)}_{HS_2}}{2}$ | $+2\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{3{g_1}^2\lambda^{(1)}_{HS_2}}{2}$ | $+2\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{15{g_1}^2\lambda^{(1)}_{HS_2}}{2}$ | $+2\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $+4\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{3{g_1}^2\lambda^{(1)}_{HS_2}}{2}$ | $+6\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $-\frac{15{g_1}^2\lambda^{(1)}_{HS_2}}{2}$ | $+6\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $-\frac{51{g_1}^2\lambda^{(1)}_{HS_2}}{2}$ | $+6\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$ | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{15{g_1}^2\lambda^{(1)}_{HS_2}}{2}$ | — | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{3{g_1}^2\lambda^{(1)}_{HS_2}}{2}$ | — | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{15{g_1}^2\lambda^{(1)}_{HS_2}}{2}$ | — | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $-\frac{51{g_1}^2\lambda^{(1)}_{HS_2}}{2}$ | — | $+6\lambda_1\lambda^{(1)}_{HS_2}$ |

**Quantity:** $\lambda^{(1)}_{HS_2}$

# Terms 4 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+4{\lambda^{(1)}_{HS_2}}^2$ |
| T2 | $+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ |
| T3 | $+6{\lambda^{(A)}_{HS_2}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | $+6{\lambda^{(A)}_{HS_2}}^2$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | $+6{\lambda^{(A)}_{HS_2}}^2$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | $+6{\lambda^{(A)}_{HS_2}}^2$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+6\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+4\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+4\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+4\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | $+6{\lambda^{(A)}_{HS_2}}^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | $+6{\lambda^{(A)}_{HS_2}}^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | $+6{\lambda^{(A)}_{HS_2}}^2$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+4{\lambda^{(1)}_{HS_2}}^2$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$ | $+6{\lambda^{(A)}_{HS_2}}^2$ |

**Quantity:** $\lambda^{(1)}_{HS_2}$

# Terms 5 of 9

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $+\frac{9{g_2}^4}{4}$ |
| T3 | $-9{g_2}^2\lambda^{(1)}_{HS_2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+\frac{9{g_2}^4}{4}$ | $-9{g_2}^2\lambda^{(1)}_{HS_2}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+\frac{9{g_2}^4}{4}$ | $-9{g_2}^2\lambda^{(1)}_{HS_2}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $\lambda^{(1)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+\frac{9{g_2}^4}{4}$ | $-9{g_2}^2\lambda^{(1)}_{HS_2}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $+\frac{9{g_2}^4}{4}$ | $-9{g_2}^2\lambda^{(1)}_{HS_2}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $+\frac{9{g_2}^4}{4}$ | $-9{g_2}^2\lambda^{(1)}_{HS_2}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $+\frac{9{g_2}^4}{4}$ | $-9{g_2}^2\lambda^{(1)}_{HS_2}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{HS_2}^{(1)}$

# Terms 6 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3{g_1}^4}{4}$ |
| T2 | $-3{g_1}^2\lambda_{HS_2}^{(1)}$ |
| T3 | $+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+\frac{3{g_1}^4}{4}$ | $-3{g_1}^2\lambda_{HS_2}^{(1)}$ | $+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\frac{3{g_1}^4}{4}$ | $-3{g_1}^2\lambda_{HS_2}^{(1)}$ | $+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+\frac{27{g_1}^4}{4}$ | $-15{g_1}^2\lambda_{HS_2}^{(1)}$ | $+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{3{g_1}^4}{4}$ | $-3{g_1}^2\lambda_{HS_2}^{(1)}$ | $+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{3{g_1}^4}{4}$ | $-3{g_1}^2\lambda_{HS_2}^{(1)}$ | $+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\frac{27{g_1}^4}{4}$ | $-15{g_1}^2\lambda_{HS_2}^{(1)}$ | $+\frac{3{\lambda_{HS_2}^{(A)}}^2}{4}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda^{(1)}_{HS_2}$

# Terms 7 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\frac{1}{2}\,\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*+3\,(\lambda_{10})^*\right)$ |
| T2 | $+\frac{1}{2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3 | $+3\lambda_{10}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\frac{1}{2}\,\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*+3\,(\lambda_{10})^*\right)$ | $+\frac{1}{2}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ | $+3\lambda_{10}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\frac{3}{4}\,\text{Tr}\left(y_1\ (y_2)^\dagger\right)+4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*+3\,(\lambda_{10})^*\right)$ | $+\frac{3}{4}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ | $+3\lambda_{10}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda^{(1)}_{HS_2}$

# Terms 8 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+24\lambda_8\,(\lambda_8)^*$ |
| T2 | $\lambda^{(1)}_{HS_2}(3T^{(2)}_\nu + 6\,\mathrm{Tr}\big(Y_d\,(Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\,(Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ |
| T3 | $\lambda^{(1)}_{HS_2}(4T^{(2)}_\nu + 6\,\mathrm{Tr}\big(Y_d\,(Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\,(Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+24\lambda_8\,(\lambda_8)^*$ | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $\lambda^{(1)}_{HS_2}(3T^{(2)}_\nu + 6\,\mathrm{Tr}\big(Y_d\,(Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\,(Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+24\lambda_8\,(\lambda_8)^*$ | $\lambda^{(1)}_{HS_2}(3T^{(2)}_\nu + 6\,\mathrm{Tr}\big(Y_d\,(Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\,(Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $\lambda^{(1)}_{HS_2}(3T^{(2)}_\nu + 6\,\mathrm{Tr}\big(Y_d\,(Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\,(Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | $\lambda^{(1)}_{HS_2}(4T^{(2)}_\nu + 6\,\mathrm{Tr}\big(Y_d\,(Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\,(Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | $\lambda^{(1)}_{HS_2}(4T^{(2)}_\nu + 6\,\mathrm{Tr}\big(Y_d\,(Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\,(Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | $\lambda^{(1)}_{HS_2}(4T^{(2)}_\nu + 6\,\mathrm{Tr}\big(Y_d\,(Y_d)^\dagger\big) + 2\,\mathrm{Tr}\big(Y_e\,(Y_e)^\dagger\big) + 6\,\mathrm{Tr}\big(Y_u\,(Y_u)^\dagger\big))$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda^{(1)}_{HS_2}$

# Terms 9 of 9

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$ |

$\beta^{(1)}_{\lambda^{(1)}_{12}}$

**Quantity:** $\lambda^{(1)}_{12}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{12}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{12}}$$

## Terms 1 of 12

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{12}(4T^{(1)}_{\nu} + 2T^{(2)}_{\nu} + 4\lambda^{(A)}_{S_2})$ |
| T2 | $-4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3 | $+4\lambda_5\,(\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda^{(1)}_{12}(4T^{(1)}_{\nu} + 2T^{(2)}_{\nu} + 4\lambda^{(A)}_{S_2})$ | $-4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | $+4\lambda_5\,(\lambda_5)^*$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(1)}_{12}(4T^{(1)}_{\nu} + 2T^{(2)}_{\nu} + 4\lambda^{(A)}_{S_2})$ | $-4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | $+4\lambda_5\,(\lambda_5)^*$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(1)}_{12}(4T^{(1)}_{\nu} + 2T^{(2)}_{\nu} + 4\lambda^{(A)}_{S_2})$ | $-4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | $+4\lambda_5\,(\lambda_5)^*$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $-2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | $+3\lambda_5\,(\lambda_5)^*$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $-2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | $+3\lambda_5\,(\lambda_5)^*$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $-2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | $+3\lambda_5\,(\lambda_5)^*$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | $+3\lambda_5\,(\lambda_5)^*$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | $+3\lambda_5\,(\lambda_5)^*$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | $+3\lambda_5\,(\lambda_5)^*$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $-4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | $+4\lambda_5\,(\lambda_5)^*$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $-4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | $+4\lambda_5\,(\lambda_5)^*$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $-4\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | $+4\lambda_5\,(\lambda_5)^*$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $-2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $-2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $-2\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ | — |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 2 of 12

| Column | Representative term structure |
|---|---|
| T1 | $-12{g_2}^2\lambda_{12}^{(1)}$ |
| T2 | $+48{g_1}^4$ |
| T3 | $-30{g_1}^2\lambda_{12}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-12{g_2}^2\lambda_{12}^{(1)}$ | $+48{g_1}^4$ | $-30{g_1}^2\lambda_{12}^{(1)}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-12{g_2}^2\lambda_{12}^{(1)}$ | — | $-6{g_1}^2\lambda_{12}^{(1)}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-12{g_2}^2\lambda_{12}^{(1)}$ | — | $-6{g_1}^2\lambda_{12}^{(1)}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-9{g_2}^2\lambda_{12}^{(1)}$ | — | $-15{g_1}^2\lambda_{12}^{(1)}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-9{g_2}^2\lambda_{12}^{(1)}$ | — | $-3{g_1}^2\lambda_{12}^{(1)}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-9{g_2}^2\lambda_{12}^{(1)}$ | — | $-15{g_1}^2\lambda_{12}^{(1)}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-9{g_2}^2\lambda_{12}^{(1)}$ | — | $-15{g_1}^2\lambda_{12}^{(1)}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-9{g_2}^2\lambda_{12}^{(1)}$ | — | $-3{g_1}^2\lambda_{12}^{(1)}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-9{g_2}^2\lambda_{12}^{(1)}$ | — | $-15{g_1}^2\lambda_{12}^{(1)}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $-12{g_2}^2\lambda_{12}^{(1)}$ | — | $-6{g_1}^2\lambda_{12}^{(1)}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $-12{g_2}^2\lambda_{12}^{(1)}$ | — | $-6{g_1}^2\lambda_{12}^{(1)}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $-12{g_2}^2\lambda_{12}^{(1)}$ | $+48{g_1}^4$ | $-30{g_1}^2\lambda_{12}^{(1)}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $+48{g_1}^4$ | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $+48{g_1}^4$ | — |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 3 of 12

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda_{12}^{(1)\,2}$ |
| T2 | $+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ |
| T3 | $+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+4\lambda_{12}^{(1)\,2}$ | $+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+4\lambda_{12}^{(1)\,2}$ | $+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+4\lambda_{12}^{(1)\,2}$ | $+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+4\lambda_{12}^{(1)\,2}$ | $+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+4\lambda_{12}^{(1)\,2}$ | $+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+4\lambda_{12}^{(1)\,2}$ | $+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+4\lambda_{12}^{(1)\,2}$ | $+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+4\lambda_{12}^{(1)\,2}$ | $+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4\lambda_{12}^{(1)\,2}$ | $+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+4\lambda_{12}^{(1)\,2}$ | $+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+4\lambda_{12}^{(1)\,2}$ | $+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+4\lambda_{12}^{(1)\,2}$ | $+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$ | $+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+4\lambda_{12}^{(1)\,2}$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+4\lambda_{12}^{(1)\,2}$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+4\lambda_{12}^{(1)\,2}$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+4\lambda_{12}^{(1)\,2}$ | — | — |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 4 of 12

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |
| T2 | $\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)})$ |
| T3 | $+\frac{9{g_2}^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | $\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)})$ | $+\frac{9{g_2}^4}{4}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | $\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)})$ | $+\frac{9{g_2}^4}{4}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | $\lambda_{12}^{(1)}(2T_\nu^{(1)}+2T_\nu^{(2)})$ | $+\frac{9{g_2}^4}{4}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | $+\frac{9{g_2}^4}{4}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | $+\frac{9{g_2}^4}{4}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | $+\frac{9{g_2}^4}{4}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ | — | — |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 5 of 12

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{27{g_1}^4}{4}$ |
| T2 | $+\frac{3{\lambda_{12}^{(A)}}^2}{4}$ |
| T3 | $+\lambda_{10}\left(\frac{3}{2}\,\mathrm{Tr}\left(y_1\ {(y_2)}^\dagger\right)+9((\lambda_{10})^*+(\lambda_{11})^*)\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+\frac{27{g_1}^4}{4}$ | $+\frac{3{\lambda_{12}^{(A)}}^2}{4}$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\frac{3{g_1}^4}{4}$ | $+\frac{3{\lambda_{12}^{(A)}}^2}{4}$ | $+\lambda_{10}\left(\frac{3}{2}\,\mathrm{Tr}\left(y_1\ {(y_2)}^\dagger\right)+9((\lambda_{10})^*+(\lambda_{11})^*)\right)$<br>$+\lambda_{11}\left(\frac{3}{2}\,\mathrm{Tr}\left(y_1\ {(y_2)}^\dagger\right)+9((\lambda_{10})^*+(\lambda_{11})^*)\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+\frac{27{g_1}^4}{4}$ | $+\frac{3{\lambda_{12}^{(A)}}^2}{4}$ | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\frac{27{g_1}^4}{4}$ | $+\frac{3{\lambda_{12}^{(A)}}^2}{4}$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{3{g_1}^4}{4}$ | $+\frac{3{\lambda_{12}^{(A)}}^2}{4}$ | $+\lambda_{10}\left(\frac{9}{4}\,\mathrm{Tr}\left(y_1\ {(y_2)}^\dagger\right)+9((\lambda_{10})^*+(\lambda_{11})^*)\right)$<br>$+\lambda_{11}\left(\frac{9}{4}\,\mathrm{Tr}\left(y_1\ {(y_2)}^\dagger\right)+9((\lambda_{10})^*+(\lambda_{11})^*)\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\frac{27{g_1}^4}{4}$ | $+\frac{3{\lambda_{12}^{(A)}}^2}{4}$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 6 of 12

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3}{2}((\lambda_{10})^{*}+(\lambda_{11})^{*})\,\mathrm{Tr}\left(y_2\ (y_1)^{\dagger}\right)$ |
| T2 | $-4\,\mathrm{Tr}\left(y_1\ (y_2)^{\dagger}\ y_2\ (y_1)^{\dagger}\right)$ |
| T3 | $+2\lambda^{\mathrm{Cross}}_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}\left(\lambda^{\mathrm{Cross}}_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}\right)^{*}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\frac{3}{2}((\lambda_{10})^{*}+(\lambda_{11})^{*})\,\mathrm{Tr}\left(y_2\ (y_1)^{\dagger}\right)$ | $-4\,\mathrm{Tr}\left(y_1\ (y_2)^{\dagger}\ y_2\ (y_1)^{\dagger}\right)$ | $+2\lambda^{\mathrm{Cross}}_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}\left(\lambda^{\mathrm{Cross}}_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}\right)^{*}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{9}{4}((\lambda_{10})^{*}+(\lambda_{11})^{*})\,\mathrm{Tr}\left(y_2\ (y_1)^{\dagger}\right)$ | $-3\,\mathrm{Tr}\left(y_1\ (y_2)^{\dagger}\ y_2\ (y_1)^{\dagger}\right)$ | $+2\lambda^{\mathrm{Cross}}_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}\left(\lambda^{\mathrm{Cross}}_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}\right)^{*}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 7 of 12

| Column | Representative term structure |
|---|---|
| T1 | $+24\lambda_9\,(\lambda_9)^*$ |
| T2 | $\lambda_{12}^{(1)}(3T_\nu^{(1)}+3T_\nu^{(2)})$ |
| T3 | $-\frac{9}{2}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+24\lambda_9\,(\lambda_9)^*$ | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $\lambda_{12}^{(1)}(3T_\nu^{(1)}+3T_\nu^{(2)})$ | $-\frac{9}{2}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+24\lambda_9\,(\lambda_9)^*$ | $\lambda_{12}^{(1)}(3T_\nu^{(1)}+3T_\nu^{(2)})$ | $-\frac{9}{2}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $\lambda_{12}^{(1)}(3T_\nu^{(1)}+3T_\nu^{(2)})$ | $-\frac{9}{2}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 8 of 12

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$ |
| T2 | $\lambda_{12}^{(1)}\left(2T_\nu^{(1)} + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right)$ |
| T3 | $+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_\nu^{(1)} + \frac{4}{3}T_\nu^{(2)}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $\lambda_{12}^{(1)}(2T_\nu^{(1)} + 4T_\nu^{(2)} + 4\lambda_{S_1}^{(A)})$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $\lambda_{12}^{(1)}\left(2T_\nu^{(1)} + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right)$ | $+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_\nu^{(1)} + \frac{4}{3}T_\nu^{(2)}\right)$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $\lambda_{12}^{(1)}\left(2T_\nu^{(1)} + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right)$ | $+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_\nu^{(1)} + \frac{4}{3}T_\nu^{(2)}\right)$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $\lambda_{12}^{(1)}\left(2T_\nu^{(1)} + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right)$ | $+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_\nu^{(1)} + \frac{4}{3}T_\nu^{(2)}\right)$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $\lambda_{12}^{(1)}\left(2T_\nu^{(1)} + 2T_\nu^{(2)} + \frac{16\lambda_{12}^{(\times)}}{3} + 4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}\right)$ | $+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_\nu^{(1)} + \frac{4}{3}T_\nu^{(2)}\right)$ |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 9 of 12

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |
| T2 | $+16{g_2}^4$ |
| T3 | $+{g_2}^2(-24\lambda_{12}^{(1)} - 16\lambda_{12}^{(\times)})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ | $+16{g_2}^4$ | $+{g_2}^2(-24\lambda_{12}^{(1)} - 16\lambda_{12}^{(\times)})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ | $+16{g_2}^4$ | $+{g_2}^2(-24\lambda_{12}^{(1)} - 16\lambda_{12}^{(\times)})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ | $+16{g_2}^4$ | $+{g_2}^2(-24\lambda_{12}^{(1)} - 16\lambda_{12}^{(\times)})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ | $+16{g_2}^4$ | $+{g_2}^2(-24\lambda_{12}^{(1)} - 16\lambda_{12}^{(\times)})$ |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 10 of 12

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2(-30\lambda_{12}^{(1)} - 20\lambda_{12}^{(\times)})$ |
| T2 | $+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)} + \frac{16\lambda_{12}^{(\times)}}{3}\right)$ |
| T3 | $+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)} + \frac{16\lambda_{12}^{(\times)}}{3}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+{g_1}^2(-30\lambda_{12}^{(1)} - 20\lambda_{12}^{(\times)})$ | $+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)} + \frac{16\lambda_{12}^{(\times)}}{3}\right)$ | $+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)} + \frac{16\lambda_{12}^{(\times)}}{3}\right)$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+{g_1}^2(-6\lambda_{12}^{(1)} - 4\lambda_{12}^{(\times)})$ | $+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)} + \frac{16\lambda_{12}^{(\times)}}{3}\right)$ | $+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)} + \frac{16\lambda_{12}^{(\times)}}{3}\right)$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+{g_1}^2(-6\lambda_{12}^{(1)} - 4\lambda_{12}^{(\times)})$ | $+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)} + \frac{16\lambda_{12}^{(\times)}}{3}\right)$ | $+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)} + \frac{16\lambda_{12}^{(\times)}}{3}\right)$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+{g_1}^2(-30\lambda_{12}^{(1)} - 20\lambda_{12}^{(\times)})$ | $+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)} + \frac{16\lambda_{12}^{(\times)}}{3}\right)$ | $+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)} + \frac{16\lambda_{12}^{(\times)}}{3}\right)$ |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 11 of 12

| Column | Representative term structure |
|---|---|
| T1 | $+16{\lambda_{12}^{(A)}}^2$ |
| T2 | $+\frac{32{\lambda_{12}^{(\times)}}^2}{3}$ |
| T3 | $+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+16{\lambda_{12}^{(A)}}^2$ | $+\frac{32{\lambda_{12}^{(\times)}}^2}{3}$ | $+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+16{\lambda_{12}^{(A)}}^2$ | $+\frac{32{\lambda_{12}^{(\times)}}^2}{3}$ | $+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+16{\lambda_{12}^{(A)}}^2$ | $+\frac{32{\lambda_{12}^{(\times)}}^2}{3}$ | $+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+16{\lambda_{12}^{(A)}}^2$ | $+\frac{32{\lambda_{12}^{(\times)}}^2}{3}$ | $+\frac{8\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}}{3}$ |

**Quantity:** $\lambda_{12}^{(1)}$

# Terms 12 of 12

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$ |

$\beta^{(1)}_{\lambda_5}$

**Quantity:** $\lambda_5$

$$16\pi^2\,\mu\frac{d\lambda_5}{d\mu} = \beta^{(1)}_{\lambda_5}$$

# Terms 1 of 6

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{HS_2})$ |
| T2 | $-\frac{21{g_2}^2\lambda_5}{2}$ |
| T3 | $-\frac{33{g_1}^2\lambda_5}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{HS_2})$ | $-\frac{21{g_2}^2\lambda_5}{2}$ | $-\frac{33{g_1}^2\lambda_5}{2}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{HS_2})$ | $-\frac{21{g_2}^2\lambda_5}{2}$ | $-\frac{9{g_1}^2\lambda_5}{2}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda_5(2T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 4\lambda^{(A)}_{HS_2})$ | $-\frac{21{g_2}^2\lambda_5}{2}$ | $-\frac{9{g_1}^2\lambda_5}{2}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $-\frac{21{g_2}^2\lambda_5}{2}$ | $-\frac{9{g_1}^2\lambda_5}{2}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{21{g_2}^2\lambda_5}{2}$ | $-\frac{9{g_1}^2\lambda_5}{2}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $-\frac{21{g_2}^2\lambda_5}{2}$ | $-\frac{33{g_1}^2\lambda_5}{2}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $-\frac{33{g_2}^2\lambda_5}{2}$ | $-\frac{33{g_1}^2\lambda_5}{2}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $-\frac{33{g_2}^2\lambda_5}{2}$ | $-\frac{9{g_1}^2\lambda_5}{2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{33{g_2}^2\lambda_5}{2}$ | $-\frac{9{g_1}^2\lambda_5}{2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $-\frac{33{g_2}^2\lambda_5}{2}$ | $-\frac{33{g_1}^2\lambda_5}{2}$ |

**Quantity:** $\lambda_5$

# Terms 2 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_{12}^{(1)}\lambda_5$ |
| T2 | $+2\lambda_1\lambda_5$ |
| T3 | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+2\lambda_{12}^{(1)}\lambda_5$ | $+2\lambda_1\lambda_5$ | $+4\lambda_{HS_1}^{(1)}\lambda_5$ |

**Quantity:** $\lambda_5$

# Terms 3 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+4\lambda_{HS_2}^{(1)}\lambda_5$ |
| T2 | $\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$ |
| T3 | $-9{g_2}^2\lambda_5$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | $\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$ | $-9{g_2}^2\lambda_5$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | $\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$ | $-9{g_2}^2\lambda_5$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | $\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$ | $-9{g_2}^2\lambda_5$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | $-9{g_2}^2\lambda_5$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | $-9{g_2}^2\lambda_5$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | $-9{g_2}^2\lambda_5$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+4\lambda_{HS_2}^{(1)}\lambda_5$ | — | — |

**Quantity:** $\lambda_5$

# Terms 4 of 6

| Column | Representative term structure |
|---|---|
| T1 | $-9{g_1}^2\lambda_5$ |
| T2 | $+2\,(\lambda_7)^*\,\mathrm{Tr}\Big(y_2\;(y_1)^\dagger\Big)$ |
| T3 | $+2\,(\lambda_8)^*\,\mathrm{Tr}\Big(y_1\;(y_2)^\dagger\Big)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $-9{g_1}^2\lambda_5$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-3{g_1}^2\lambda_5$ | $+2\,(\lambda_7)^*\,\mathrm{Tr}\Big(y_2\;(y_1)^\dagger\Big)$ | $+2\,(\lambda_8)^*\,\mathrm{Tr}\Big(y_1\;(y_2)^\dagger\Big)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $-9{g_1}^2\lambda_5$ | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $-9{g_1}^2\lambda_5$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-3{g_1}^2\lambda_5$ | $+3\,(\lambda_7)^*\,\mathrm{Tr}\Big(y_2\;(y_1)^\dagger\Big)$ | $+3\,(\lambda_8)^*\,\mathrm{Tr}\Big(y_1\;(y_2)^\dagger\Big)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $-9{g_1}^2\lambda_5$ | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_5$

# Terms 5 of 6

| Column | Representative term structure |
|---|---|
| T1 | $-4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\,(\lambda_7)^*$ |
| T2 | $+4\lambda_{11}\,(\lambda_7)^*$ |
| T3 | $+4\,(\lambda_8)^*\,((\lambda_{10})^* - \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\,(\lambda_7)^*$ | $+4\lambda_{11}\,(\lambda_7)^*$ | $+4\,(\lambda_8)^*\,((\lambda_{10})^* - \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\,(\lambda_7)^*$ | $+4\lambda_{11}\,(\lambda_7)^*$ | $+4\,(\lambda_8)^*\,((\lambda_{10})^* - \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_5$

# Terms 6 of 6

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_5 \left( \frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} \right)$ |
| T2 | $\lambda_5 (T_\nu^{(1)} + 2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$ |
| T3 | $\lambda_5 \left( T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\lambda_5 \left( \frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} \right)$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $\lambda_5 \left( \frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} \right)$ | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\lambda_5 \left( \frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} \right)$ | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $\lambda_5 (T_\nu^{(1)} + 2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $\lambda_5 (T_\nu^{(1)} + 2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $\lambda_5 (T_\nu^{(1)} + 2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - 4\lambda_{HS_1}^{(A)})$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | $\lambda_5 \left( T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right)$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | $\lambda_5 \left( T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right)$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | $\lambda_5 \left( T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right)$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | $\lambda_5 \left( T_\nu^{(1)} + T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3} \right)$ |

$\beta^{(1)}_{\lambda_{12}^{(A)}}$

**Quantity:** $\lambda_{12}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{(A)}}{d\mu} = \beta^{(1)}_{\lambda_{12}^{(A)}}$$

## Terms 1 of 8

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$ |
| T2 | $-\frac{8}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ |
| T3 | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$ | $-\frac{8}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$ | $-\frac{8}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $\lambda_{12}^{(A)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$ | $-\frac{8}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $-\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $-\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $-\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $+\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $-\frac{2}{9}\lambda_5\,(\lambda_5)^*$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $+\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $-\frac{2}{9}\lambda_5\,(\lambda_5)^*$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $+\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $-\frac{2}{9}\lambda_5\,(\lambda_5)^*$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $+\frac{2}{3}\,\mathrm{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $-\frac{2}{9}\lambda_5\,(\lambda_5)^*$ |

**Quantity:** $\lambda_{12}^{(A)}$

# Terms 2 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$ |
| T2 | $-9{g_2}^2\lambda_{12}^{(A)}$ |
| T3 | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$ | $-9{g_2}^2\lambda_{12}^{(A)}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $-9{g_2}^2\lambda_{12}^{(A)}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$ | $-9{g_2}^2\lambda_{12}^{(A)}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$ | $-9{g_2}^2\lambda_{12}^{(A)}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $-9{g_2}^2\lambda_{12}^{(A)}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+{g_1}^2\left(18{g_2}^2-15\lambda_{12}^{(A)}\right)$ | $-9{g_2}^2\lambda_{12}^{(A)}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+{g_1}^2\left(16{g_2}^2-30\lambda_{12}^{(A)}\right)$ | $-24{g_2}^2\lambda_{12}^{(A)}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $-24{g_2}^2\lambda_{12}^{(A)}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-24{g_2}^2\lambda_{12}^{(A)}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+{g_1}^2\left(16{g_2}^2-30\lambda_{12}^{(A)}\right)$ | $-24{g_2}^2\lambda_{12}^{(A)}$ | $+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$ |

**Quantity:** $\lambda_{12}^{(A)}$

# Terms 3 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ |
| T2 | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ |
| T3 | $+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ | $+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ | $+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ | $+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ | $+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ | $+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ | $+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$ | $+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$ | — |

**Quantity:** $\lambda_{12}^{(A)}$

# Terms 4 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda_{10}\left(\frac{2}{3}\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)+\frac{4}{3}(5\,(\lambda_{10})^*+(\lambda_{11})^*)\right)$ |
| T2 | $+\lambda_{11}\left(\frac{2}{3}\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)+\frac{4}{3}((\lambda_{10})^*+5\,(\lambda_{11})^*)\right)$ |
| T3 | $+\frac{2}{3}((\lambda_{10})^*+(\lambda_{11})^*)\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\lambda_{10}\left(\frac{2}{3}\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)+\frac{4}{3}(5\,(\lambda_{10})^*+(\lambda_{11})^*)\right)$ | $+\lambda_{11}\left(\frac{2}{3}\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)+\frac{4}{3}((\lambda_{10})^*+5\,(\lambda_{11})^*)\right)$ | $+\frac{2}{3}((\lambda_{10})^*+(\lambda_{11})^*)\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{12}^{(A)}$

# Terms 5 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{64}{3}\lambda_9\,(\lambda_9)^*$ |
| T2 | $+{g_1}^2\left(-6{g_2}^2-3\lambda_{12}^{(A)}\right)$ |
| T3 | $\lambda_{12}^{(A)}(3T_\nu^{(1)}+3T_\nu^{(2)})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\frac{64}{3}\lambda_9\,(\lambda_9)^*$ | $+{g_1}^2\left(-6{g_2}^2-3\lambda_{12}^{(A)}\right)$ | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | $\lambda_{12}^{(A)}(3T_\nu^{(1)}+3T_\nu^{(2)})$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{64}{3}\lambda_9\,(\lambda_9)^*$ | $+{g_1}^2\left(-6{g_2}^2-3\lambda_{12}^{(A)}\right)$ | $\lambda_{12}^{(A)}(3T_\nu^{(1)}+3T_\nu^{(2)})$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | $\lambda_{12}^{(A)}(3T_\nu^{(1)}+3T_\nu^{(2)})$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{12}^{(A)}$

# Terms 6 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda_{10}\left(\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)+\frac{4}{3}(5\,(\lambda_{10})^*+(\lambda_{11})^*)\right)$ |
| T2 | $+\lambda_{11}\left(\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)+\frac{4}{3}((\lambda_{10})^*+5\,(\lambda_{11})^*)\right)$ |
| T3 | $+((\lambda_{10})^*+(\lambda_{11})^*)\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\lambda_{10}\left(\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)+\frac{4}{3}(5\,(\lambda_{10})^*+(\lambda_{11})^*)\right)$ | $+\lambda_{11}\left(\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)+\frac{4}{3}((\lambda_{10})^*+5\,(\lambda_{11})^*)\right)$ | $+((\lambda_{10})^*+(\lambda_{11})^*)\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{12}^{(A)}$

# Terms 7 of 8

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big)$ |
| T2 | $\lambda_{12}^{(A)}\left(2T_\nu^{(1)}+2T_\nu^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$ |
| T3 | $+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-\frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big)$ | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $\lambda_{12}^{(A)}\left(2T_\nu^{(1)}+2T_\nu^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$ | $+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $\lambda_{12}^{(A)}\left(2T_\nu^{(1)}+2T_\nu^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$ | $+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $\lambda_{12}^{(A)}\left(2T_\nu^{(1)}+2T_\nu^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$ | $+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $\lambda_{12}^{(A)}\left(2T_\nu^{(1)}+2T_\nu^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$ | $+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$ |

**Quantity:** $\lambda_{12}^{(A)}$

# Terms 8 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$ |
| T2 | $+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3 | $-6{g_1}^2\lambda_{12}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$ | $+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$ | $+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ | $-6{g_1}^2\lambda_{12}^{(A)}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$ | $+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ | $-6{g_1}^2\lambda_{12}^{(A)}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$ | $+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$ | — |

$\beta^{(1)}_{\lambda_{12}^{(\times)}}$

**Quantity:** $\lambda_{12}^{(\times)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{(\times)}}{d\mu} = \beta^{(1)}_{\lambda_{12}^{(\times)}}$$

## Terms 1 of 6

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_{12}^{(1)}\left(\frac{1}{2}T_\nu^{(1)} + \frac{1}{2}T_\nu^{(2)} + 8\lambda_{12}^{(\times)} + \lambda_{S_1}^{(A)} + \lambda_{S_2}^{(A)}\right)$ |
| T2 | $+\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$ |
| T3 | $-\frac{7}{6}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda_{12}^{(1)}\left(\frac{1}{2}T_\nu^{(1)} + \frac{1}{2}T_\nu^{(2)} + 8\lambda_{12}^{(\times)} + \lambda_{S_1}^{(A)} + \lambda_{S_2}^{(A)}\right)$ | $+\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$ | $-\frac{7}{6}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda_{12}^{(1)}\left(\frac{1}{2}T_\nu^{(1)} + \frac{1}{2}T_\nu^{(2)} + 8\lambda_{12}^{(\times)} + \lambda_{S_1}^{(A)} + \lambda_{S_2}^{(A)}\right)$ | $+\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$ | $-\frac{7}{6}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda_{12}^{(1)}\left(\frac{1}{2}T_\nu^{(1)} + \frac{1}{2}T_\nu^{(2)} + 8\lambda_{12}^{(\times)} + \lambda_{S_1}^{(A)} + \lambda_{S_2}^{(A)}\right)$ | $+\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$ | $-\frac{7}{6}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $\lambda_{12}^{(1)}\left(\frac{1}{2}T_\nu^{(1)} + \frac{1}{2}T_\nu^{(2)} + 8\lambda_{12}^{(\times)} + \lambda_{S_1}^{(A)} + \lambda_{S_2}^{(A)}\right)$ | $+\lambda_{12}^{(\times)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$ | $-\frac{7}{6}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big)$ |

**Quantity:** $\lambda_{12}^{(\times)}$

# Terms 2 of 6

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{1}{2}\lambda_5\,(\lambda_5)^*$ |
| T2 | $+\frac{33{g_2}^4}{4}$ |
| T3 | $+{g_2}^2(-6\lambda_{12}^{(1)} - 24\lambda_{12}^{(\times)})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{1}{2}\lambda_5\,(\lambda_5)^*$ | $+\frac{33{g_2}^4}{4}$ | $+{g_2}^2(-6\lambda_{12}^{(1)} - 24\lambda_{12}^{(\times)})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{1}{2}\lambda_5\,(\lambda_5)^*$ | $+\frac{33{g_2}^4}{4}$ | $+{g_2}^2(-6\lambda_{12}^{(1)} - 24\lambda_{12}^{(\times)})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{1}{2}\lambda_5\,(\lambda_5)^*$ | $+\frac{33{g_2}^4}{4}$ | $+{g_2}^2(-6\lambda_{12}^{(1)} - 24\lambda_{12}^{(\times)})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $-\frac{1}{2}\lambda_5\,(\lambda_5)^*$ | $+\frac{33{g_2}^4}{4}$ | $+{g_2}^2(-6\lambda_{12}^{(1)} - 24\lambda_{12}^{(\times)})$ |

**Quantity:** $\lambda_{12}^{(\times)}$

# Terms 3 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+12{g_1}^4$ |
| T2 | $+{g_1}^2\left(-\frac{15\lambda_{12}^{(1)}}{2}-30\lambda_{12}^{(\times)}\right)$ |
| T3 | $+{\lambda_{12}^{(1)}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+12{g_1}^4$ | $+{g_1}^2\left(-\frac{15\lambda_{12}^{(1)}}{2}-30\lambda_{12}^{(\times)}\right)$ | $+{\lambda_{12}^{(1)}}^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $+{g_1}^2\left(-\frac{3\lambda_{12}^{(1)}}{2}-6\lambda_{12}^{(\times)}\right)$ | $+{\lambda_{12}^{(1)}}^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $+{g_1}^2\left(-\frac{3\lambda_{12}^{(1)}}{2}-6\lambda_{12}^{(\times)}\right)$ | $+{\lambda_{12}^{(1)}}^2$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+12{g_1}^4$ | $+{g_1}^2\left(-\frac{15\lambda_{12}^{(1)}}{2}-30\lambda_{12}^{(\times)}\right)$ | $+{\lambda_{12}^{(1)}}^2$ |

**Quantity:** $\lambda_{12}^{(\times)}$

# Terms 4 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$ |
| T2 | $+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$ |
| T3 | $+4\lambda_{12}^{(A)\,2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$ | $+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$ | $+4\lambda_{12}^{(A)\,2}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$ | $+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$ | $+4\lambda_{12}^{(A)\,2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$ | $+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$ | $+4\lambda_{12}^{(A)\,2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$ | $+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$ | $+4\lambda_{12}^{(A)\,2}$ |

**Quantity:** $\lambda_{12}^{(\times)}$

# Terms 5 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+11\lambda_{12}^{(\times)\,2}$ |
| T2 | $-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$ |
| T3 | $-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+11\lambda_{12}^{(\times)\,2}$ | $-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$ | $-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+11\lambda_{12}^{(\times)\,2}$ | $-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$ | $-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+11\lambda_{12}^{(\times)\,2}$ | $-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$ | $-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+11\lambda_{12}^{(\times)\,2}$ | $-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$ | $-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$ |

**Quantity:** $\lambda_{12}^{(\times)}$

# Terms 6 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$ |

$\beta^{(1)}_{\lambda^{(A)}_{HS_1}}$

**Quantity:** $\lambda^{(A)}_{HS_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_1}}$$

## Terms 1 of 8

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $+\frac{8}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3 | $+\frac{16}{3}\lambda_5\,(\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+\frac{8}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+\frac{16}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+\frac{8}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+\frac{16}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $\lambda^{(A)}_{HS_1}(2T^{(1)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+\frac{8}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+\frac{16}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | $-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+\frac{16}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+\frac{16}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+\frac{16}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | $-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $-\frac{8}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $-\frac{8}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $-\frac{8}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $-\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $-\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $-\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $-\frac{4}{3}\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $-\frac{4}{3}\lambda_5\,(\lambda_5)^*$ |

**Quantity:** $\lambda_{HS_1}^{(A)}$

# Terms 2 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(-18{g_2}^2-15\lambda_{HS_1}^{(A)}\right)$ |
| T2 | $-9{g_2}^2\lambda_{HS_1}^{(A)}$ |
| T3 | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+{g_1}^2\left(-18{g_2}^2-15\lambda_{HS_1}^{(A)}\right)$ | $-9{g_2}^2\lambda_{HS_1}^{(A)}$ | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_1}^{(A)}\right)$ | $-9{g_2}^2\lambda_{HS_1}^{(A)}$ | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $-9{g_2}^2\lambda_{HS_1}^{(A)}$ | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+{g_1}^2\left(-18{g_2}^2-15\lambda_{HS_1}^{(A)}\right)$ | $-9{g_2}^2\lambda_{HS_1}^{(A)}$ | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+{g_1}^2\left(-6{g_2}^2-3\lambda_{HS_1}^{(A)}\right)$ | $-9{g_2}^2\lambda_{HS_1}^{(A)}$ | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $-9{g_2}^2\lambda_{HS_1}^{(A)}$ | $+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{HS_1}^{(A)}$

# Terms 3 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_1\lambda_{HS_1}^{(A)}$ |
| T2 | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ |
| T3 | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+2\lambda_1\lambda_{HS_1}^{(A)}$ | $+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$ | $+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$ |

**Quantity:** $\lambda^{(A)}_{HS_1}$

# Terms 4 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(-\frac{2}{3}\,\text{Tr}\left(y_1\ (y_2)^\dagger\right)-\frac{4}{3}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*+(\lambda_{11})^*)\right)$ |
| T2 | $-\frac{2}{3}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\,\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3 | $+\frac{64}{3}\lambda_7\,(\lambda_7)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(-\frac{2}{3}\,\text{Tr}\left(y_1\ (y_2)^\dagger\right)-\frac{4}{3}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*+(\lambda_{11})^*)\right)$ | $-\frac{2}{3}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\,\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ | $+\frac{64}{3}\lambda_7\,(\lambda_7)^*$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | $+\frac{64}{3}\lambda_7\,(\lambda_7)^*$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda^{(A)}_{HS_1}$

# Terms 5 of 8

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{4}{3}\lambda_{11}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ |
| T2 | $+{g_1}^2\left(6{g_2}^2-3\lambda^{(A)}_{HS_1}\right)$ |
| T3 | $\lambda^{(A)}_{HS_1}(3T^{(1)}_\nu+6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-\frac{4}{3}\lambda_{11}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $+{g_1}^2\left(6{g_2}^2-3\lambda^{(A)}_{HS_1}\right)$ | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | $\lambda^{(A)}_{HS_1}(3T^{(1)}_\nu+6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-\frac{4}{3}\lambda_{11}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ | — | $\lambda^{(A)}_{HS_1}(3T^{(1)}_\nu+6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | $+{g_1}^2\left(6{g_2}^2-3\lambda^{(A)}_{HS_1}\right)$ | $\lambda^{(A)}_{HS_1}(3T^{(1)}_\nu+6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda^{(A)}_{HS_1}$

# Terms 6 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(-\operatorname{Tr}\left(y_1\ (y_2)^\dagger\right)-\frac{4}{3}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*+(\lambda_{11})^*)\right)$ |
| T2 | $-\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\operatorname{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3 | $\lambda^{(A)}_{HS_1}(2T^{(1)}_\nu+6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda^{(A)}_{S_1})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(-\operatorname{Tr}\left(y_1\ (y_2)^\dagger\right)-\frac{4}{3}(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*+(\lambda_{11})^*)\right)$ | $-\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\operatorname{Tr}\left(y_2\ (y_1)^\dagger\right)$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | $\lambda^{(A)}_{HS_1}(2T^{(1)}_\nu+6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda^{(A)}_{S_1})$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | $\lambda^{(A)}_{HS_1}(2T^{(1)}_\nu+6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda^{(A)}_{S_1})$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | $\lambda^{(A)}_{HS_1}(2T^{(1)}_\nu+6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda^{(A)}_{S_1})$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | $\lambda^{(A)}_{HS_1}(2T^{(1)}_\nu+6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda^{(A)}_{S_1})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | $\lambda^{(A)}_{HS_1}(2T^{(1)}_\nu+6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda^{(A)}_{S_1})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | $\lambda^{(A)}_{HS_1}(2T^{(1)}_\nu+6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda^{(A)}_{S_1})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | $\lambda^{(A)}_{HS_1}(2T^{(1)}_\nu+6\operatorname{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\operatorname{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\operatorname{Tr}\left(Y_u\ (Y_u)^\dagger\right)+16\lambda^{(A)}_{S_1})$ |

**Quantity:** $\lambda_{HS_1}^{(A)}$

# Terms 7 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(-4{g_2}^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$ |
| T2 | $-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$ |
| T3 | $-\frac{3{g_1}^2\lambda_{HS_1}^{(A)}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+{g_1}^2\left(-4{g_2}^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$ | $-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$ | $-\frac{3{g_1}^2\lambda_{HS_1}^{(A)}}{2}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | $-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+{g_1}^2\left(-8{g_2}^2-\frac{51\lambda_{HS_1}^{(A)}}{2}\right)$ | $-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$ | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+{g_1}^2\left(-4{g_2}^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$ | $-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$ | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$ | $-\frac{3{g_1}^2\lambda_{HS_1}^{(A)}}{2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $-\frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}$ | — |

**Quantity:** $\lambda_{HS_1}^{(A)}$

# Terms 8 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left({4{g_2}^2} - \frac{15\lambda_{HS_1}^{(A)}}{2}\right)$ |
| T2 | $+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+{g_1}^2\left({4{g_2}^2} - \frac{15\lambda_{HS_1}^{(A)}}{2}\right)$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+{g_1}^2\left({4{g_2}^2} - \frac{15\lambda_{HS_1}^{(A)}}{2}\right)$ | $+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$ |

$\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$

**Quantity:** $\lambda^{(A)}_{HS_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{HS_2}}$$

## Terms 1 of 8

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$ |
| T2 | $+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3 | $+\frac{8}{3}\lambda_5\,(\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$ | $+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+\frac{8}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$ | $+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+\frac{8}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$ | $+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $+\frac{8}{3}\lambda_5\,(\lambda_5)^*$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$ | $+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $-\frac{8}{9}\lambda_5\,(\lambda_5)^*$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$ | $+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $-\frac{8}{9}\lambda_5\,(\lambda_5)^*$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$ | $+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $-\frac{8}{9}\lambda_5\,(\lambda_5)^*$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right) + 16\lambda^{(A)}_{S_2})$ | $+\frac{4}{3}\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ | $-\frac{8}{9}\lambda_5\,(\lambda_5)^*$ |

**Quantity:** $\lambda_{HS_2}^{(A)}$

# Terms 2 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(-4{g_2}^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$ |
| T2 | $-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$ |
| T3 | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+{g_1}^2\left(-4{g_2}^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$ | $-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$ | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$ | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$ | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+{g_1}^2\left(-4{g_2}^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$ | $-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$ | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$ | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$ | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$ | $+2\lambda_1\lambda_{HS_2}^{(A)}$ |

**Quantity:** $\lambda^{(A)}_{HS_2}$

# Terms 3 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ |
| T2 | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ |
| T3 | $-\frac{3{g_1}^2\lambda^{(A)}_{HS_2}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | $-\frac{3{g_1}^2\lambda^{(A)}_{HS_2}}{2}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | $-\frac{3{g_1}^2\lambda^{(A)}_{HS_2}}{2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+8\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$ | $+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$ | — |

**Quantity:** $\lambda^{(A)}_{HS_2}$

# Terms 4 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(4{g_2}^2-\frac{15\lambda^{(A)}_{HS_2}}{2}\right)$ |
| T2 | $\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T3 | $+8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+{g_1}^2\left(4{g_2}^2-\frac{15\lambda^{(A)}_{HS_2}}{2}\right)$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | $\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | $\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | $\lambda^{(A)}_{HS_2}(2T^{(2)}_{\nu}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+8\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | $-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | $-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | $-4\,\mathrm{Tr}\left(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+{g_1}^2\left(4{g_2}^2-\frac{15\lambda^{(A)}_{HS_2}}{2}\right)$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+{g_1}^2\left(8{g_2}^2-\frac{51\lambda^{(A)}_{HS_2}}{2}\right)$ | — | — |

**Quantity:** $\lambda_{HS_2}^{(A)}$

# Terms 5 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+16\lambda_5 \left(\lambda_5\right)^*$ |
| T2 | $+{g_1}^2 \left(-6{g_2}^2 - 3\lambda_{HS_2}^{(A)}\right)$ |
| T3 | $-9{g_2}^2\lambda_{HS_2}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+16\lambda_5 \left(\lambda_5\right)^*$ | $+{g_1}^2 \left(-6{g_2}^2 - 3\lambda_{HS_2}^{(A)}\right)$ | $-9{g_2}^2\lambda_{HS_2}^{(A)}$ |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+16\lambda_5 \left(\lambda_5\right)^*$ | — | $-9{g_2}^2\lambda_{HS_2}^{(A)}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+16\lambda_5 \left(\lambda_5\right)^*$ | — | $-9{g_2}^2\lambda_{HS_2}^{(A)}$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+16\lambda_5 \left(\lambda_5\right)^*$ | $+{g_1}^2 \left(-6{g_2}^2 - 3\lambda_{HS_2}^{(A)}\right)$ | $-9{g_2}^2\lambda_{HS_2}^{(A)}$ |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+16\lambda_5 \left(\lambda_5\right)^*$ | — | $-9{g_2}^2\lambda_{HS_2}^{(A)}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+16\lambda_5 \left(\lambda_5\right)^*$ | — | $-9{g_2}^2\lambda_{HS_2}^{(A)}$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda^{(A)}_{HS_2}$

# Terms 6 of 8

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$ |
| T2 | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(-2\,\text{Tr}\left(y_1\ (y_2)^\dagger\right) - 4(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*))$ |
| T3 | $-2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | $+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$ | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(-2\,\text{Tr}\left(y_1\ (y_2)^\dagger\right) - 4(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*))$ | $-2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | $+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$ | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$ | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(-3\,\text{Tr}\left(y_1\ (y_2)^\dagger\right) - 4(2\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* + (\lambda_{10})^*))$ | $-3\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $+\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$ | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{HS_2}^{(A)}$

# Terms 7 of 8

| Column | Representative term structure |
|---|---|
| T1 | $-4\lambda_{10}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ |
| T2 | $+64\lambda_8\,(\lambda_8)^*$ |
| T3 | $+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-4\lambda_{10}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ | $+64\lambda_8\,(\lambda_8)^*$ | $+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | $+{g_1}^2\left(18{g_2}^2-15\lambda_{HS_2}^{(A)}\right)$ |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-4\lambda_{10}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ | $+64\lambda_8\,(\lambda_8)^*$ | $+{g_1}^2\left(6{g_2}^2-3\lambda_{HS_2}^{(A)}\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | $+{g_1}^2\left(18{g_2}^2-15\lambda_{HS_2}^{(A)}\right)$ |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{HS_2}^{(A)}$

# Terms 8 of 8

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_{HS_2}^{(A)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | $\lambda_{HS_2}^{(A)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $\lambda_{HS_2}^{(A)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | $\lambda_{HS_2}^{(A)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | $+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | $+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | $+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$ |

$\beta^{(1)}_{\lambda_7}$

**Quantity:** $\lambda_7$

$$16\pi^2\,\mu\frac{d\lambda_7}{d\mu} = \beta^{(1)}_{\lambda_7}$$

# Terms 1 of 5

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_7(2T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $+ (\lambda_5)^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3 | $-2\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\,(\lambda_5)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\lambda_7(2T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+ (\lambda_5)^*\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ | $-2\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\,(\lambda_5)^*$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | $-2\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\,(\lambda_5)^*$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_7$

# Terms 2 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_{11}\,(\lambda_5)^*$ |
| T2 | $-9{g_2}^2\lambda_7$ |
| T3 | $-3{g_1}^2\lambda_7$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+2\lambda_{11}\,(\lambda_5)^*$ | $-9{g_2}^2\lambda_7$ | $-3{g_1}^2\lambda_7$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+2\lambda_{11}\,(\lambda_5)^*$ | $-9{g_2}^2\lambda_7$ | $-3{g_1}^2\lambda_7$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_7$

# Terms 3 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_1\lambda_7$ |
| T2 | $+8\lambda_{HS_1}^{(1)}\lambda_7$ |
| T3 | $+\dfrac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+2\lambda_1\lambda_7$ | $+8\lambda_{HS_1}^{(1)}\lambda_7$ | $+\dfrac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+2\lambda_1\lambda_7$ | $+8\lambda_{HS_1}^{(1)}\lambda_7$ | $+\dfrac{4\lambda_{HS_1}^{(A)}\lambda_7}{3}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_7$

# Terms 4 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_7\lambda_{S_1}^{(1)}$ |
| T2 | $+4\lambda_8\lambda_9$ |
| T3 | $\lambda_7(3T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+2\lambda_7\lambda_{S_1}^{(1)}$ | $+4\lambda_8\lambda_9$ | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+2\lambda_7\lambda_{S_1}^{(1)}$ | $+4\lambda_8\lambda_9$ | $\lambda_7(3T_\nu^{(1)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_7$

# Terms 5 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3}{2}\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_2\;(y_1)^\dagger\Big)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 |
|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{3}{2}\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_2\;(y_1)^\dagger\Big)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — |

$\beta^{(1)}_{\lambda_{12}}$

**Quantity:** $\lambda_{12}$

$$16\pi^2\,\mu\frac{d\lambda_{12}}{d\mu} = \beta^{(1)}_{\lambda_{12}}$$

## Terms 1 of 5

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{HS_1}\left(\frac{1}{2}\,\mathrm{Tr}\big(y_2\ (y_1)^\dagger\big) + 4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 3\lambda_{11}\right)$ |
| T2 | $+\lambda^{(A)}_{HS_1}\left(-\frac{1}{8}\,\mathrm{Tr}\big(y_2\ (y_1)^\dagger\big) - \frac{\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2} - \frac{\lambda_{11}}{4}\right)$ |
| T3 | $+\lambda^{(1)}_{HS_2}\left(\frac{1}{2}\,\mathrm{Tr}\big(y_2\ (y_1)^\dagger\big) + 4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 3\lambda_{10}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\lambda^{(1)}_{HS_1}\left(\frac{1}{2}\,\mathrm{Tr}\big(y_2\ (y_1)^\dagger\big) + 4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 3\lambda_{11}\right)$ | $+\lambda^{(A)}_{HS_1}\left(-\frac{1}{8}\,\mathrm{Tr}\big(y_2\ (y_1)^\dagger\big) - \frac{\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2} - \frac{\lambda_{11}}{4}\right)$ | $+\lambda^{(1)}_{HS_2}\left(\frac{1}{2}\,\mathrm{Tr}\big(y_2\ (y_1)^\dagger\big) + 4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 3\lambda_{10}\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $\lambda^{(1)}_{HS_1}\left(\frac{3}{4}\,\mathrm{Tr}\big(y_2\ (y_1)^\dagger\big) + 4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 3\lambda_{11}\right)$ | $+\lambda^{(A)}_{HS_1}\left(-\frac{3}{16}\,\mathrm{Tr}\big(y_2\ (y_1)^\dagger\big) - \frac{\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2} - \frac{\lambda_{11}}{4}\right)$ | $+\lambda^{(1)}_{HS_2}\left(\frac{3}{4}\,\mathrm{Tr}\big(y_2\ (y_1)^\dagger\big) + 4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 3\lambda_{10}\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{12}$

# Terms 2 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda^{(A)}_{HS_2}\left(-\frac{3}{8}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)-\frac{3\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{3\lambda_{10}}{4}\right)$ |
| T2 | $+\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T3 | $-4\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\lambda^{(A)}_{HS_2}\left(-\frac{3}{8}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)-\frac{3\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{3\lambda_{10}}{4}\right)$ | $+\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(T_\nu^{(1)}+T_\nu^{(2)}+6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $-4\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\lambda^{(A)}_{HS_2}\left(-\frac{9}{16}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)-\frac{3\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{3\lambda_{10}}{4}\right)$ | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{12}$

# Terms 3 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+6\lambda_9 \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ |
| T2 | $-6\,(\lambda_8)^*\,(\lambda_5)^*$ |
| T3 | $-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+6\lambda_9 \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ | $-6\,(\lambda_8)^*\,(\lambda_5)^*$ | $-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+6\lambda_9 \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*$ | $-6\,(\lambda_8)^*\,(\lambda_5)^*$ | $-9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

# Terms 4 of 5

| Column | Representative term structure |
|---|---|
| T1 | $-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |
| T2 | $+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |
| T3 | $+3\lambda_1\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ | $+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ | $+3\lambda_1\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ | $+2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ | $+3\lambda_1\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{12}$

# Terms 5 of 5

| Column | Representative term structure |
|---|---|
| T1 | $-6\lambda_7\lambda_5$ |
| T2 | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-6\lambda_7\lambda_5$ | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-6\lambda_7\lambda_5$ | $+\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\,\text{Tr}\left(Y_d\ (Y_d)^\dagger\right)+2\,\text{Tr}\left(Y_e\ (Y_e)^\dagger\right)+6\,\text{Tr}\left(Y_u\ (Y_u)^\dagger\right)\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — |

$\beta^{(1)}_{\lambda_8}$

**Quantity:** $\lambda_8$

$$16\pi^2\,\mu\frac{d\lambda_8}{d\mu} = \beta^{(1)}_{\lambda_8}$$

# Terms 1 of 4

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_8(2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T2 | $+(\lambda_5)^*\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)$ |
| T3 | $+4\lambda_7\,(\lambda_9)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\lambda_8(2T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+(\lambda_5)^*\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)$ | $+4\lambda_7\,(\lambda_9)^*$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | $+4\lambda_7\,(\lambda_9)^*$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_8$

# Terms 2 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+2\,(\lambda_5)^*\,((\lambda_{10})^* - \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$ |
| T2 | $-9{g_2}^2\lambda_8$ |
| T3 | $-3{g_1}^2\lambda_8$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+2\,(\lambda_5)^*\,((\lambda_{10})^* - \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$ | $-9{g_2}^2\lambda_8$ | $-3{g_1}^2\lambda_8$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+2\,(\lambda_5)^*\,((\lambda_{10})^* - \left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*)$ | $-9{g_2}^2\lambda_8$ | $-3{g_1}^2\lambda_8$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_8$

# Terms 3 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_1\lambda_8$ |
| T2 | $+8\lambda^{(1)}_{HS_2}\lambda_8$ |
| T3 | $+4\lambda^{(A)}_{HS_2}\lambda_8$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+2\lambda_1\lambda_8$ | $+8\lambda^{(1)}_{HS_2}\lambda_8$ | $+4\lambda^{(A)}_{HS_2}\lambda_8$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+2\lambda_1\lambda_8$ | $+8\lambda^{(1)}_{HS_2}\lambda_8$ | $+4\lambda^{(A)}_{HS_2}\lambda_8$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_8$

# Terms 4 of 4

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda_8 \lambda_{S_2}^{(1)}$ |
| T2 | $\lambda_8(3T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ |
| T3 | $+\frac{3}{2}\,(\lambda_5)^*\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+2\lambda_8 \lambda_{S_2}^{(1)}$ | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+2\lambda_8 \lambda_{S_2}^{(1)}$ | $\lambda_8(3T_\nu^{(2)} + 6\,\mathrm{Tr}\left(Y_d\ (Y_d)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ (Y_e)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ (Y_u)^\dagger\right))$ | $+\frac{3}{2}\,(\lambda_5)^*\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

$\beta^{(1)}_{\lambda^{(A)}_{S_1}}$

**Quantity:** $\lambda^{(A)}_{S_1}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{S_1}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{S_1}}$$

## Terms 1 of 6

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{S_1}\left(\frac{1}{2}T^{(1)}_{\nu} + 13\lambda^{(A)}_{S_1}\right)$ |
| T2 | $+4\lambda^{(A)}_{S_1}T^{(1)}_{\nu}$ |
| T3 | $-\operatorname{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(1)}_{S_1}\left(\frac{1}{2}T^{(1)}_{\nu} + 13\lambda^{(A)}_{S_1}\right)$ | $+4\lambda^{(A)}_{S_1}T^{(1)}_{\nu}$ | $-\operatorname{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(1)}_{S_1}\left(\frac{1}{2}T^{(1)}_{\nu} + 13\lambda^{(A)}_{S_1}\right)$ | $+4\lambda^{(A)}_{S_1}T^{(1)}_{\nu}$ | $-\operatorname{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $\lambda^{(1)}_{S_1}\left(\frac{1}{2}T^{(1)}_{\nu} + 13\lambda^{(A)}_{S_1}\right)$ | $+4\lambda^{(A)}_{S_1}T^{(1)}_{\nu}$ | $-\operatorname{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda^{(1)}_{S_1}\left(\frac{1}{2}T^{(1)}_{\nu} + 13\lambda^{(A)}_{S_1}\right)$ | $+4\lambda^{(A)}_{S_1}T^{(1)}_{\nu}$ | $-\operatorname{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(1)}_{S_1}\left(\frac{1}{2}T^{(1)}_{\nu} + 13\lambda^{(A)}_{S_1}\right)$ | $+4\lambda^{(A)}_{S_1}T^{(1)}_{\nu}$ | $-\operatorname{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(1)}_{S_1}\left(\frac{1}{2}T^{(1)}_{\nu} + 13\lambda^{(A)}_{S_1}\right)$ | $+4\lambda^{(A)}_{S_1}T^{(1)}_{\nu}$ | $-\operatorname{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $\lambda^{(1)}_{S_1}\left(\frac{1}{2}T^{(1)}_{\nu} + 13\lambda^{(A)}_{S_1}\right)$ | $+4\lambda^{(A)}_{S_1}T^{(1)}_{\nu}$ | $-\operatorname{Tr}\left(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\right)$ |

**Quantity:** $\lambda_{S_1}^{(A)}$

# Terms 2 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3{g_2}^4}{4}$ |
| T2 | $+{g_1}^2\left(9{g_2}^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$ |
| T3 | $+{g_2}^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3{g_2}^4}{4}$ | $+{g_1}^2\left(9{g_2}^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$ | $+{g_2}^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3{g_2}^4}{4}$ | — | $+{g_2}^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+\frac{3{g_2}^4}{4}$ | $+{g_1}^2\left(9{g_2}^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$ | $+{g_2}^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{3{g_2}^4}{4}$ | — | $+{g_2}^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3{g_2}^4}{4}$ | $+{g_1}^2\left(9{g_2}^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$ | $+{g_2}^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3{g_2}^4}{4}$ | — | $+{g_2}^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\frac{3{g_2}^4}{4}$ | $+{g_1}^2\left(9{g_2}^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$ | $+{g_2}^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$ |

**Quantity:** $\lambda_{S_1}^{(A)}$

# Terms 3 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3{g_1}^4}{2}$ |
| T2 | $+\frac{{\lambda_{12}^{(1)}}^2}{4}$ |
| T3 | $+\frac{{\lambda_{HS_1}^{(1)}}^2}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3{g_1}^4}{2}$ | $+\frac{{\lambda_{12}^{(1)}}^2}{4}$ | $+\frac{{\lambda_{HS_1}^{(1)}}^2}{2}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | $+\frac{{\lambda_{12}^{(1)}}^2}{4}$ | $+\frac{{\lambda_{HS_1}^{(1)}}^2}{2}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+\frac{3{g_1}^4}{2}$ | $+\frac{{\lambda_{12}^{(1)}}^2}{4}$ | $+\frac{{\lambda_{HS_1}^{(1)}}^2}{2}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | $+\frac{{\lambda_{HS_1}^{(1)}}^2}{2}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3{g_1}^4}{2}$ | — | $+\frac{{\lambda_{HS_1}^{(1)}}^2}{2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | $+\frac{{\lambda_{HS_1}^{(1)}}^2}{2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\frac{3{g_1}^4}{2}$ | — | $+\frac{{\lambda_{HS_1}^{(1)}}^2}{2}$ |

**Quantity:** $\lambda_{S_1}^{(A)}$

# Terms 4 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+3\lambda_{HS_1}^{(A)\,2}$ |
| T2 | $+\dfrac{7\lambda_{S_1}^{(1)\,2}}{4}$ |
| T3 | $+22\lambda_{S_1}^{(A)\,2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | $+3\lambda_{HS_1}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_1}^{(1)\,2}}{4}$ | $+22\lambda_{S_1}^{(A)\,2}$ |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | $+3\lambda_{HS_1}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_1}^{(1)\,2}}{4}$ | $+22\lambda_{S_1}^{(A)\,2}$ |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | $+3\lambda_{HS_1}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_1}^{(1)\,2}}{4}$ | $+22\lambda_{S_1}^{(A)\,2}$ |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+3\lambda_{HS_1}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_1}^{(1)\,2}}{4}$ | $+22\lambda_{S_1}^{(A)\,2}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+3\lambda_{HS_1}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_1}^{(1)\,2}}{4}$ | $+22\lambda_{S_1}^{(A)\,2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+3\lambda_{HS_1}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_1}^{(1)\,2}}{4}$ | $+22\lambda_{S_1}^{(A)\,2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+3\lambda_{HS_1}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_1}^{(1)\,2}}{4}$ | $+22\lambda_{S_1}^{(A)\,2}$ |

**Quantity:** $\lambda_{S_1}^{(A)}$

# Terms 5 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+{g_1}^2\left(36{g_2}^2-6\lambda_{S_1}^{(1)}-48\lambda_{S_1}^{(A)}\right)$ |
| T2 | $+24{g_1}^4$ |
| T3 | $+\frac{3{\lambda_{12}^{(1)}}^2}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+{g_1}^2\left(36{g_2}^2-6\lambda_{S_1}^{(1)}-48\lambda_{S_1}^{(A)}\right)$ | $+24{g_1}^4$ | $+\frac{3{\lambda_{12}^{(1)}}^2}{4}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | $+\frac{3{\lambda_{12}^{(1)}}^2}{4}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | $+\frac{3{\lambda_{12}^{(1)}}^2}{4}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | $+\frac{3{\lambda_{12}^{(1)}}^2}{4}$ |

**Quantity:** $\lambda_{S_1}^{(A)}$

# Terms 6 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ |
| T2 | $+12\lambda_{12}^{(A)\,2}$ |
| T3 | $-\frac{\lambda_{12}^{(\times)\,2}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ | $+12\lambda_{12}^{(A)\,2}$ | $-\frac{\lambda_{12}^{(\times)\,2}}{2}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ | $+12\lambda_{12}^{(A)\,2}$ | $-\frac{\lambda_{12}^{(\times)\,2}}{2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ | $+12\lambda_{12}^{(A)\,2}$ | $-\frac{\lambda_{12}^{(\times)\,2}}{2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ | $+12\lambda_{12}^{(A)\,2}$ | $-\frac{\lambda_{12}^{(\times)\,2}}{2}$ |

$\beta^{(1)}_{\lambda_{11}}$

**Quantity:** $\lambda_{11}$

$$16\pi^2\,\mu\frac{d\lambda_{11}}{d\mu} = \beta^{(1)}_{\lambda_{11}}$$

## Terms 1 of 5

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_{12}^{(1)}(\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + 6\lambda_{10} + 6\lambda_{11})$ |
| T2 | $+\lambda_{12}^{(A)}\left(\frac{1}{12}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + \frac{\lambda_{10}}{6} + \frac{5\lambda_{11}}{6}\right)$ |
| T3 | $+\lambda_{S_1}^{(1)}(\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + 12\lambda_{11})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\lambda_{12}^{(1)}(\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + 6\lambda_{10} + 6\lambda_{11})$ | $+\lambda_{12}^{(A)}\left(\frac{1}{12}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + \frac{\lambda_{10}}{6} + \frac{5\lambda_{11}}{6}\right)$ | $+\lambda_{S_1}^{(1)}(\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + 12\lambda_{11})$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $+\lambda_{12}^{(A)}\left(\frac{1}{8}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + \frac{\lambda_{10}}{6} + \frac{5\lambda_{11}}{6}\right)$ | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{11}$

# Terms 2 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda_9(2\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right) + 4((\lambda_{10})^* + 5\,(\lambda_{11})^*))$ |
| T2 | $+\lambda_{11}(3T_\nu^{(1)} + T_\nu^{(2)})$ |
| T3 | $-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\lambda_9(2\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right) + 4((\lambda_{10})^* + 5\,(\lambda_{11})^*))$ | $+\lambda_{11}(3T_\nu^{(1)} + T_\nu^{(2)})$ | $-4\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\lambda_9(3\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right) + 4((\lambda_{10})^* + 5\,(\lambda_{11})^*))$ | — | $-5\,\mathrm{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

# Terms 3 of 5

| Column | Representative term structure |
|---|---|
| T1 | $-9{g_2}^2\lambda_{11}$ |
| T2 | $-3{g_1}^2\lambda_{11}$ |
| T3 | $+2\lambda^{(1)}_{HS_1}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-9{g_2}^2\lambda_{11}$ | $-3{g_1}^2\lambda_{11}$ | $+2\lambda^{(1)}_{HS_1}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-9{g_2}^2\lambda_{11}$ | $-3{g_1}^2\lambda_{11}$ | $+2\lambda^{(1)}_{HS_1}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{11}$

# Terms 4 of 5

| Column | Representative term structure |
|---|---|
| T1 | $-\dfrac{\lambda_{HS_1}^{(A)}\,\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{6}$ |
| T2 | $+4\lambda_7\lambda_5$ |
| T3 | $\lambda_{12}^{(1)}\left(\dfrac{3}{2}\,\mathrm{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{10}+6\lambda_{11}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-\dfrac{\lambda_{HS_1}^{(A)}\,\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{6}$ | $+4\lambda_7\lambda_5$ | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-\dfrac{\lambda_{HS_1}^{(A)}\,\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{6}$ | $+4\lambda_7\lambda_5$ | $\lambda_{12}^{(1)}\left(\dfrac{3}{2}\,\mathrm{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{10}+6\lambda_{11}\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{11}$

# Terms 5 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda_{S_1}^{(1)}\left(\frac{3}{2}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)+12\lambda_{11}\right)$ |
| T2 | $+\lambda_{11}\left(\frac{9}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\lambda_{S_1}^{(1)}\left(\frac{3}{2}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)+12\lambda_{11}\right)$ | $+\lambda_{11}\left(\frac{9}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — |

$\beta^{(1)}_{\lambda_9}$

**Quantity:** $\lambda_9$

$$16\pi^2\,\mu\frac{d\lambda_9}{d\mu} = \beta^{(1)}_{\lambda_9}$$

# Terms 1 of 6

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_9(2T_\nu^{(1)} + 2T_\nu^{(2)})$ |
| T2 | $+\lambda_{10}(\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + 2\lambda_{11})$ |
| T3 | $+\lambda_{11}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\lambda_9(2T_\nu^{(1)} + 2T_\nu^{(2)})$ | $+\lambda_{10}(\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + 2\lambda_{11})$ | $+\lambda_{11}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_9$

# Terms 2 of 6

| Column | Representative term structure |
|---|---|
| T1 | $-2\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\right)$ |
| T2 | $+4\lambda_7\ (\lambda_8)^*$ |
| T3 | $-9{g_2}^2\lambda_9$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-2\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\right)$ | $+4\lambda_7\ (\lambda_8)^*$ | $-9{g_2}^2\lambda_9$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $+4\lambda_7\ (\lambda_8)^*$ | $-9{g_2}^2\lambda_9$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_9$

# Terms 3 of 6

| Column | Representative term structure |
|---|---|
| T1 | $-3{g_1}^2\lambda_9$ |
| T2 | $+8\lambda_{12}^{(1)}\lambda_9$ |
| T3 | $+\frac{4\lambda_{12}^{(A)}\lambda_9}{3}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $-3{g_1}^2\lambda_9$ | $+8\lambda_{12}^{(1)}\lambda_9$ | $+\frac{4\lambda_{12}^{(A)}\lambda_9}{3}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $-3{g_1}^2\lambda_9$ | $+8\lambda_{12}^{(1)}\lambda_9$ | $+\frac{4\lambda_{12}^{(A)}\lambda_9}{3}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_9$

# Terms 4 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}{}^2$ |
| T2 | $+2\lambda^{(1)}_{S_1}\lambda_9$ |
| T3 | $+2\lambda_9\lambda^{(1)}_{S_2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}{}^2$ | $+2\lambda^{(1)}_{S_1}\lambda_9$ | $+2\lambda_9\lambda^{(1)}_{S_2}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}{}^2$ | $+2\lambda^{(1)}_{S_1}\lambda_9$ | $+2\lambda_9\lambda^{(1)}_{S_2}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_9$

# Terms 5 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+5{\lambda_{10}}^2$ |
| T2 | $\lambda_9(3T_\nu^{(1)} + 3T_\nu^{(2)})$ |
| T3 | $+\lambda_{10}\left(\frac{3}{2}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + 2\lambda_{11}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+5{\lambda_{10}}^2$<br>$+5{\lambda_{11}}^2$ | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+5{\lambda_{10}}^2$<br>$+5{\lambda_{11}}^2$ | $\lambda_9(3T_\nu^{(1)} + 3T_\nu^{(2)})$ | $+\lambda_{10}\left(\frac{3}{2}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + 2\lambda_{11}\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_9$

# Terms 6 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3}{2}\lambda_{11}\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\Big)$ |
| T2 | $-\frac{5}{2}\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\Big)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\frac{3}{2}\lambda_{11}\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\Big)$ | $-\frac{5}{2}\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\Big)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — |

$\beta^{(1)}_{\lambda_{10}}$

**Quantity:** $\lambda_{10}$

$$16\pi^2\,\mu\frac{d\lambda_{10}}{d\mu} = \beta^{(1)}_{\lambda_{10}}$$

# Terms 1 of 5

| Column | Representative term structure |
|---|---|
| T1 | $\lambda_{12}^{(1)}(\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + 6\lambda_{10} + 6\lambda_{11})$ |
| T2 | $+\lambda_{12}^{(A)}\left(\frac{1}{12}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + \frac{5\lambda_{10}}{6} + \frac{\lambda_{11}}{6}\right)$ |
| T3 | $+\lambda_9(2\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right) + 4(5\,(\lambda_{10})^* + (\lambda_{11})^*))$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $\lambda_{12}^{(1)}(\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + 6\lambda_{10} + 6\lambda_{11})$ | $+\lambda_{12}^{(A)}\left(\frac{1}{12}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + \frac{5\lambda_{10}}{6} + \frac{\lambda_{11}}{6}\right)$ | $+\lambda_9(2\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right) + 4(5\,(\lambda_{10})^* + (\lambda_{11})^*))$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | $+\lambda_{12}^{(A)}\left(\frac{1}{8}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + \frac{5\lambda_{10}}{6} + \frac{\lambda_{11}}{6}\right)$ | $+\lambda_9(3\,\mathrm{Tr}\left(y_1\ (y_2)^\dagger\right) + 4(5\,(\lambda_{10})^* + (\lambda_{11})^*))$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{10}$

# Terms 2 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda_{S_2}^{(1)}(\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + 12\lambda_{10})$ |
| T2 | $+\lambda_{10}(T_\nu^{(1)} + 3T_\nu^{(2)})$ |
| T3 | $-4\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+\lambda_{S_2}^{(1)}(\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right) + 12\lambda_{10})$ | $+\lambda_{10}(T_\nu^{(1)} + 3T_\nu^{(2)})$ | $-4\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | $-5\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{10}$

# Terms 3 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+4\,(\lambda_8)^*\,(\lambda_5)^*$ |
| T2 | $-9{g_2}^2\lambda_{10}$ |
| T3 | $-3{g_1}^2\lambda_{10}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+4\,(\lambda_8)^*\,(\lambda_5)^*$ | $-9{g_2}^2\lambda_{10}$ | $-3{g_1}^2\lambda_{10}$ |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+4\,(\lambda_8)^*\,(\lambda_5)^*$ | $-9{g_2}^2\lambda_{10}$ | $-3{g_1}^2\lambda_{10}$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{10}$

# Terms 4 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+2\lambda^{(1)}_{HS_2}\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ |
| T2 | $-\dfrac{\lambda^{(A)}_{HS_2}\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}$ |
| T3 | $\lambda^{(1)}_{12}\left(\dfrac{3}{2}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | $+2\lambda^{(1)}_{HS_2}\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ | $-\dfrac{\lambda^{(A)}_{HS_2}\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}$ | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+2\lambda^{(1)}_{HS_2}\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$ | $-\dfrac{\lambda^{(A)}_{HS_2}\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}$ | $\lambda^{(1)}_{12}\left(\dfrac{3}{2}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | — |

**Quantity:** $\lambda_{10}$

# Terms 5 of 5

| Column | Representative term structure |
|---|---|
| T1 | $+\lambda_{S_2}^{(1)}\left(\frac{3}{2}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)+12\lambda_{10}\right)$ |
| T2 | $+\lambda_{10}\left(\frac{3}{2}T_\nu^{(1)}+\frac{9}{2}T_\nu^{(2)}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 |
|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | $+\lambda_{S_2}^{(1)}\left(\frac{3}{2}\,\mathrm{Tr}\left(y_2\ (y_1)^\dagger\right)+12\lambda_{10}\right)$ | $+\lambda_{10}\left(\frac{3}{2}T_\nu^{(1)}+\frac{9}{2}T_\nu^{(2)}\right)$ |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — |

$\beta^{(1)}_{\lambda^{(A)}_{S_2}}$

**Quantity:** $\lambda^{(A)}_{S_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(A)}_{S_2}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{S_2}}$$

## Terms 1 of 6

| Column | Representative term structure |
|---|---|
| T1 | $\lambda^{(1)}_{S_2}\left(\frac{1}{2}T^{(2)}_{\nu} + 13\lambda^{(A)}_{S_2}\right)$ |
| T2 | $+4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ |
| T3 | $-\operatorname{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda^{(1)}_{S_2}\left(\frac{1}{2}T^{(2)}_{\nu} + 13\lambda^{(A)}_{S_2}\right)$ | $+4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-\operatorname{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(1)}_{S_2}\left(\frac{1}{2}T^{(2)}_{\nu} + 13\lambda^{(A)}_{S_2}\right)$ | $+4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-\operatorname{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(1)}_{S_2}\left(\frac{1}{2}T^{(2)}_{\nu} + 13\lambda^{(A)}_{S_2}\right)$ | $+4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-\operatorname{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $\lambda^{(1)}_{S_2}\left(\frac{1}{2}T^{(2)}_{\nu} + 13\lambda^{(A)}_{S_2}\right)$ | $+4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-\operatorname{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $\lambda^{(1)}_{S_2}\left(\frac{1}{2}T^{(2)}_{\nu} + 13\lambda^{(A)}_{S_2}\right)$ | $+4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-\operatorname{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $\lambda^{(1)}_{S_2}\left(\frac{1}{2}T^{(2)}_{\nu} + 13\lambda^{(A)}_{S_2}\right)$ | $+4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-\operatorname{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $\lambda^{(1)}_{S_2}\left(\frac{1}{2}T^{(2)}_{\nu} + 13\lambda^{(A)}_{S_2}\right)$ | $+4\lambda^{(A)}_{S_2}T^{(2)}_{\nu}$ | $-\operatorname{Tr}\left(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\right)$ |

**Quantity:** $\lambda_{S_2}^{(A)}$

# Terms 2 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3{g_2}^4}{4}$ |
| T2 | $+{g_1}^2\left(9{g_2}^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$ |
| T3 | $+{g_2}^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{3{g_2}^4}{4}$ | $+{g_1}^2\left(9{g_2}^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$ | $+{g_2}^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3{g_2}^4}{4}$ | — | $+{g_2}^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3{g_2}^4}{4}$ | $+{g_1}^2\left(9{g_2}^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$ | $+{g_2}^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{3{g_2}^4}{4}$ | $+{g_1}^2\left(9{g_2}^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$ | $+{g_2}^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3{g_2}^4}{4}$ | — | $+{g_2}^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3{g_2}^4}{4}$ | $+{g_1}^2\left(9{g_2}^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$ | $+{g_2}^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\frac{3{g_2}^4}{4}$ | — | $+{g_2}^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$ |

**Quantity:** $\lambda_{S_2}^{(A)}$

# Terms 3 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3{g_1}^4}{2}$ |
| T2 | $+\frac{{\lambda_{12}^{(1)}}^2}{4}$ |
| T3 | $+\frac{{\lambda_{HS_2}^{(1)}}^2}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{3{g_1}^4}{2}$ | $+\frac{{\lambda_{12}^{(1)}}^2}{4}$ | $+\frac{{\lambda_{HS_2}^{(1)}}^2}{2}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | $+\frac{{\lambda_{12}^{(1)}}^2}{4}$ | $+\frac{{\lambda_{HS_2}^{(1)}}^2}{2}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3{g_1}^4}{2}$ | $+\frac{{\lambda_{12}^{(1)}}^2}{4}$ | $+\frac{{\lambda_{HS_2}^{(1)}}^2}{2}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{3{g_1}^4}{2}$ | — | $+\frac{{\lambda_{HS_2}^{(1)}}^2}{2}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | $+\frac{{\lambda_{HS_2}^{(1)}}^2}{2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3{g_1}^4}{2}$ | — | $+\frac{{\lambda_{HS_2}^{(1)}}^2}{2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | — | — | $+\frac{{\lambda_{HS_2}^{(1)}}^2}{2}$ |

**Quantity:** $\lambda_{S_2}^{(A)}$

# Terms 4 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+3\lambda_{HS_2}^{(A)\,2}$ |
| T2 | $+\dfrac{7\lambda_{S_2}^{(1)\,2}}{4}$ |
| T3 | $+22\lambda_{S_2}^{(A)\,2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+3\lambda_{HS_2}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_2}^{(1)\,2}}{4}$ | $+22\lambda_{S_2}^{(A)\,2}$ |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+3\lambda_{HS_2}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_2}^{(1)\,2}}{4}$ | $+22\lambda_{S_2}^{(A)\,2}$ |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+3\lambda_{HS_2}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_2}^{(1)\,2}}{4}$ | $+22\lambda_{S_2}^{(A)\,2}$ |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+3\lambda_{HS_2}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_2}^{(1)\,2}}{4}$ | $+22\lambda_{S_2}^{(A)\,2}$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+3\lambda_{HS_2}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_2}^{(1)\,2}}{4}$ | $+22\lambda_{S_2}^{(A)\,2}$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+3\lambda_{HS_2}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_2}^{(1)\,2}}{4}$ | $+22\lambda_{S_2}^{(A)\,2}$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+3\lambda_{HS_2}^{(A)\,2}$ | $+\dfrac{7\lambda_{S_2}^{(1)\,2}}{4}$ | $+22\lambda_{S_2}^{(A)\,2}$ |

**Quantity:** $\lambda_{S_2}^{(A)}$

# Terms 5 of 6

| Column | Representative term structure |
|---|---|
| T1 | $+\frac{3{\lambda_{12}^{(1)}}^2}{4}$ |
| T2 | $+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ |
| T3 | $+12{\lambda_{12}^{(A)}}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $+\frac{3{\lambda_{12}^{(1)}}^2}{4}$ | $+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ | $+12{\lambda_{12}^{(A)}}^2$ |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $+\frac{3{\lambda_{12}^{(1)}}^2}{4}$ | $+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ | $+12{\lambda_{12}^{(A)}}^2$ |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $+\frac{3{\lambda_{12}^{(1)}}^2}{4}$ | $+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ | $+12{\lambda_{12}^{(A)}}^2$ |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $+\frac{3{\lambda_{12}^{(1)}}^2}{4}$ | $+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$ | $+12{\lambda_{12}^{(A)}}^2$ |

**Quantity:** $\lambda_{S_2}^{(A)}$

# Terms 6 of 6

| Column | Representative term structure |
|---|---|
| T1 | $-\frac{{\lambda_{12}^{(\times)}}^2}{2}$ |
| T2 | $+{g_1}^2\left(36{g_2}^2-6\lambda_{S_2}^{(1)}-48\lambda_{S_2}^{(A)}\right)$ |
| T3 | $+24{g_1}^4$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$ | T1 | T2 | T3 |
|---|---|---|---|---|---|---|---|---|---|---|
| T3-A-m4 | $-4$ | 1 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | — | — | — |
| T3-A-m2 | $-2$ | 1 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-A-p0 | 0 | 1 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-B-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 1 | $-1$ | — | — | — |
| T3-B-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 1 | 0 | — | — | — |
| T3-B-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 1 | 1 | — | — | — |
| T3-C-m3 | $-3$ | 2 | $-\frac{3}{2}$ | 2 | $-\frac{1}{2}$ | 3 | $-1$ | — | — | — |
| T3-C-m1 | $-1$ | 2 | $-\frac{1}{2}$ | 2 | $\frac{1}{2}$ | 3 | 0 | — | — | — |
| T3-C-p1 | 1 | 2 | $\frac{1}{2}$ | 2 | $\frac{3}{2}$ | 3 | 1 | — | — | — |
| T3-D-m2 | $-2$ | 3 | $-1$ | 1 | 0 | 2 | $-\frac{1}{2}$ | — | — | — |
| T3-D-p0 | 0 | 3 | 0 | 1 | 1 | 2 | $\frac{1}{2}$ | — | — | — |
| T3-D-p2 | 2 | 3 | 1 | 1 | 2 | 2 | $\frac{3}{2}$ | — | — | — |
| T3-E-m4 | $-4$ | 3 | $-2$ | 3 | $-1$ | 2 | $-\frac{3}{2}$ | $-\frac{{\lambda_{12}^{(\times)}}^2}{2}$ | — | — |
| T3-E-m2 | $-2$ | 3 | $-1$ | 3 | 0 | 2 | $-\frac{1}{2}$ | $-\frac{{\lambda_{12}^{(\times)}}^2}{2}$ | — | — |
| T3-E-p0 | 0 | 3 | 0 | 3 | 1 | 2 | $\frac{1}{2}$ | $-\frac{{\lambda_{12}^{(\times)}}^2}{2}$ | — | — |
| T3-E-p2 | 2 | 3 | 1 | 3 | 2 | 2 | $\frac{3}{2}$ | $-\frac{{\lambda_{12}^{(\times)}}^2}{2}$ | $+{g_1}^2\left(36{g_2}^2-6\lambda_{S_2}^{(1)}-48\lambda_{S_2}^{(A)}\right)$ | $+24{g_1}^4$ |